

Optimal searches for a moving target on paths and grids

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Abstract

An invisible target starts at an unknown vertex of a finite graph. Each day a searcher inspects at most m vertices simultaneously; after a miss, the target must traverse exactly one edge. We study the minimum number of days that guarantees capture. We give exact formulas for every daily budget on paths and on rectangular grids with two, three, or four rows. We develop a comparison theorem that compresses entire search strategies while preserving their daily budgets. It yields exact frontier recurrences on cylinders of odd length and on equal-sided boxes in every dimension. For every box with all side lengths odd, in arbitrary dimension, we prove bipartite open-neighborhood isoperimetric nesting. This reduces the unrestricted game to two counts and gives an exact algorithm polynomial in the number of rooms. We determine the profiles explicitly for odd rectangles and prove a three-probe time formula on every five-row board with at least five columns, as well as every budget on every even-length five-row board. At the minimum budget, odd rectangles of each odd width from three through fifteen have an exact time affine in their length; a proved interval-insertion argument extends finite rational certificates to every odd length. More generally, for any fixed all-odd cross-section and any fixed feasible budget, capture time on sufficiently long odd cylinders has a proved affine period, with an explicit threshold and a finite procedure determining the remaining constants. We also determine every budget on the $3 \times 3 \times 3$ box, prove a five-inspection feasibility threshold for every $3 \times 3 \times n$ box with $n \geq 3$, and establish the exact five-inspection time $18n - 36$ for every odd $n \geq 3$. The path, two-row, and $3 \times 3 \times 3$ classifications have complete Lean proofs with actual-walk semantics. The account reconstructs the author's October 2019–March 2020 path work and records the subsequent AI-assisted proof development. General closed formulas for arbitrary rectangular boxes remain open in this investigation.

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1 Introduction and results

A princess moves through a building every night. Each morning a searcher may open several doors. The princess knows the search plan, may start anywhere, and must move to an adjacent room after each unsuccessful day. The problem is to guarantee capture as quickly as possible. Even on a single corridor the answer has a parity correction: dividing the amount of uncertainty by the apparent daily progress is not always enough.

This is a minimum-time version of the Hunters and Rabbit game. A path P_n has n vertices, and $P_{n_1} \square \cdots \square P_{n_d}$ denotes the Cartesian box with those side lengths. Write $T_m(G)$ for the optimal guaranteed number of inspection days, with value ∞ when no finite guarantee exists. Section 2 specifies the order of inspection and compulsory movement precisely.

1.1 The exact narrow-grid formulas

On a path, $T_m(P_n) = 1$ if $n \leq m$. With one inspection per day, $T_1(P_2) = 2$ and $T_1(P_n) = 2n - 4$ for $n \geq 3$. For $n > m \geq 2$,

$$T_m(P_n) = \left\lceil \frac{2n - 4}{2m - 1} \right\rceil + \varepsilon, \quad (1)$$

where $\varepsilon = 1$ exactly when $n \equiv m + 1 \pmod{2m - 1}$ and n, m are not both even. This theorem is the domain-corrected result of the author's original 2019–2020 work. We give a shorter construction and a matching-based lower proof, and formalize the complete statement in Lean.

For two rows and $n \geq 2$, one daily inspection is insufficient, and $m \geq 2n$ gives one day. In the remaining range put $N = n - 1$, $M = m - 1$. Then

$$T_m(P_2 \square P_n) = \left\lceil \frac{2N}{M} \right\rceil + \mathbf{1}_{\{M \text{ even and } N \equiv M/2 \pmod{M}\}}. \quad (2)$$

For three rows and $n \geq 2$, the one-day threshold is $3n$, the two-day threshold is $\lfloor 3n/2 \rfloor$, and one inspection is insufficient. For $2 \leq m < \lfloor 3n/2 \rfloor$, write $3n - 4 = q(2m - 3) + r$, $0 \leq r < 2m - 3$. The answer is

$$\begin{cases} 2q, & r = 0, \\ 2q + 1, & r > 0 \text{ and } m \geq r + 4 - \mathbf{1}_{\{n \text{ even, } q \text{ odd}\}}, \\ 2q + 2, & \text{otherwise.} \end{cases} \quad (3)$$

For four rows and $n \geq 4$, the one-day threshold is $4n$, the two-day threshold is $2n$, and budgets at most two are insufficient. For $3 \leq m < 2n$, write $2n - 2 = q(m - 2) + r$, $0 \leq r < m - 2$. The answer is

$$\begin{cases} 2q, & r = 0, \\ 2q + 1, & r = 1 \text{ or } (r \geq 2 \text{ and } m \geq 2r + 4), \\ 2q + 2, & \text{otherwise.} \end{cases} \quad (4)$$

Rotating a shorter four-row board reduces it to a preceding family. In particular the minimum feasible budgets give $T_2(P_3 \square P_n) = 6n - 8$ and $T_3(P_4 \square P_n) = 4n - 4$ in their stated nondegenerate ranges. For five rows and $n \geq 5$, the minimum feasible budget is three, and

$$T_3(P_5 \square P_n) = 10n - 20. \quad (5)$$

The even-length proof uses a finite symbolic boundary certificate valid for every length, together with a history-dependent potential. The odd-length proof follows from the explicit neighborhood

profiles. For every even $n \geq 6$, putting $h = 5n/2$ also gives

$$T_4(P_5 \square P_n) = 2 \left\lceil \frac{2h - 8}{3} \right\rceil.$$

Theorem 10.5 gives a closed quotient-and-remainder formula for every larger budget on these even-length boards. For odd lengths, Corollary 4.4 gives the exact two-count recurrence at every budget, including an attaining inspection schedule. Together with rotation of the shorter boards, these determine feasibility, optimal time, and an optimal strategy for every five-row board; the odd-length all-budget answer is an exact recurrence rather than a general closed expression.

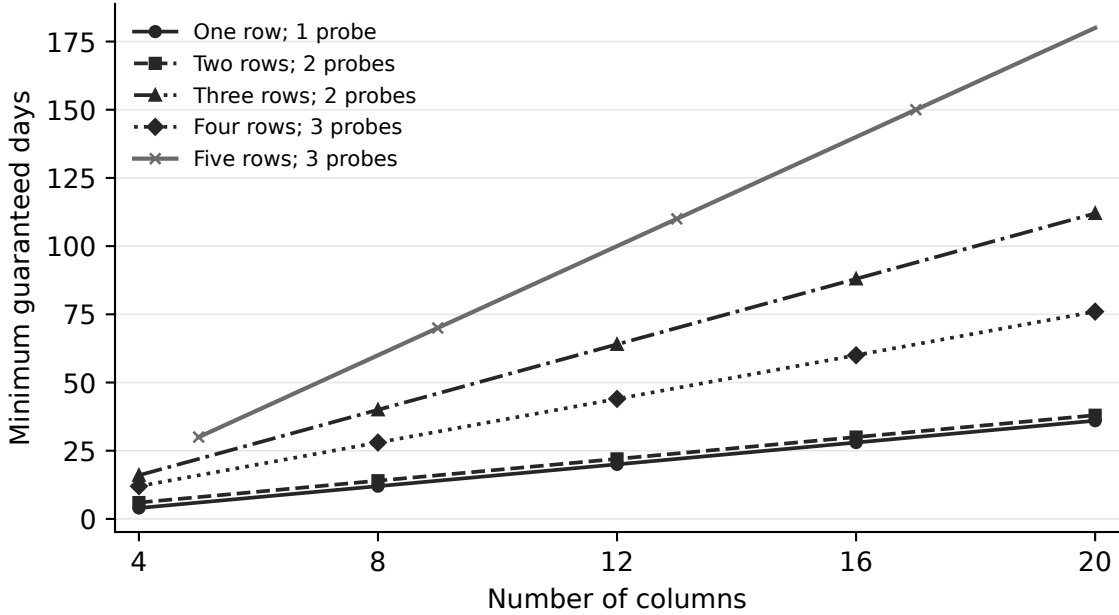


Figure 1: Optimal time at the smallest feasible daily budget for four narrow families and the five-row three-probe theorem. Each line is a proved formula, not a fit to search data.

1.2 The structural results

The common difficulty is temporal. A small possible-position set may have a particularly favorable next neighborhood, but that neighborhood need not contain an equally favorable set for the following day. We retain a history bit where necessary, and also prove a more general comparison theorem. A monotone, cardinality-preserving set map C with

$$N(C(A)) \subseteq C(N(A))$$

compresses a whole strategy without increasing any day's inspection budget. Idempotent maps supply an exact optimal-strategy normal form. This assertion is stronger than a one-step isoperimetric inequality.

We construct such operators from odd paths and square diagonal shifts. They give exact recurrences on smaller families of states in arbitrary dimension. More strongly, on every box whose side lengths are all odd, we prove that the restricted simplicial orders minimize open neighborhoods

and that neighborhoods of prefixes are prefixes. An induction using fiber compressions and a weighted exchange lemma reduces the unrestricted game to two cardinalities. The resulting exact algorithm is polynomial in the number of rooms and returns an optimal schedule. A recurrence remains distinct from a closed expression: the general box problem is not declared solved by the existence of an exponential search over all possible-position sets.

The three-dimensional example has a particularly concise answer:

m	0:4	5	6	7:8	9:11	12	13:26	27 or more
$T_m(P_3^3)$	∞	18	10	6	4	3	2	1.

Its complete lower and upper proofs are checked in Lean. A separate height construction proves $h(P_3 \square P_3 \square P_n) = 5$ for $n \geq 3$, where $h(G)$ is the minimum feasible daily budget.

1.3 Prior work and provenance

Britnell and Wildon [6] studied the princess puzzle with one daily inspection, including the path value $2n - 4$. Haslegrave studied the associated evasion game [8]; related moving- target search appears in Beluhov and Kolev [2]. Abramovskaya, Fomin, Golovach, and Pilipczuk [1] developed Hunters and Rabbit results, including the rectangular-grid feasibility threshold $\lfloor \min(a, b)/2 \rfloor + 1$. Bolkema and Groothuis proved isoperimetric nesting and the hunting number for hypercubes [5]. We use precise versions of these results; minimum-time claims do not follow from feasibility alone. Parity-restricted boundary minimization on the binary cube goes back to Körner and Wei [11]. Compression proofs and local-to-global principles have a substantial earlier literature, including Bezrukov and Serra [4]; their closed-neighborhood theorem is distinct from the open-neighborhood argument given here.

Dmitry Kamenetsky recorded conjectural two- and three-inspection path formulas in March 2018 [9, 10]. Equation (1) proves those special cases and extends them to arbitrary budgets. The author’s independent path work took place between October 2019 and March 2020 and was written in two Bulgarian conference manuscripts [13, 14]. The one-probe restriction omitted from a printed general formula is made explicit here. The original conference documents are preserved.

The time-budget inverse parameter is also considered in recent work [3]; we do not claim the general optimization question as new. Recontamination can matter substantially on general graphs [7], so a presumed monotone search is never used as an unproved lower-bound assumption. Classical closed-neighborhood isoperimetry for Cartesian grids [12] supplies one of our higher-dimensional applications. Our use of that theorem is distinguished from the open-neighborhood argument for odd rectangles.

Remark 1.1 (Status of this manuscript). This is a research manuscript in preparation, not an arXiv submission. The ordinary theorems have internal independent proof reviews; exact Lean coverage is stated in Section 14. Attribution of the 2026 extensions is still undergoing a focused literature review. The manuscript makes no blanket claim that every derived ingredient is new to the literature.

2 The game and its possible positions

Let $G = (V, E)$ be a finite simple graph without isolated vertices, and let $m \geq 1$ be the daily inspection budget. Every connected board with at least two rooms satisfies this graph assumption;

a one-room board takes one inspection and is treated separately. The target chooses an unknown initial vertex. On day $t = 1, 2, \dots$, the searcher inspects a set $S_t \subseteq V$ with $|S_t| \leq m$. If the target occupies a vertex of S_t , it is caught. Otherwise it must move along exactly one edge before the next day. Inspections within a day are simultaneous. The searcher receives no information about an unsuccessful inspection other than the miss.

Write $T_m(G)$ for the minimum number of days in which capture can be guaranteed, and put $T_m(G) = \infty$ if there is no finite guarantee. The associated feasibility threshold is

$$h(G) = \min\{m \geq 1 : T_m(G) < \infty\}.$$

We consider deterministic guarantees. Before capture, every observation is a miss. Thus an adaptive strategy has only one continuing observation history and is represented by a sequence $(S_t)_{t \geq 1}$. This sequence formulation does not restrict the searcher's guaranteed performance.

For $A \subseteq V$, its open neighborhood is

$$N(A) = \{v \in V : \text{some } u \in A \text{ satisfies } uv \in E\}.$$

Define the possible-position sets immediately before inspection by

$$B_1 = V, \quad B_{t+1} = N(B_t \setminus S_t).$$

The following elementary interpretation is used in every subsequent argument.

Lemma 2.1. *For each $t \geq 1$, a vertex v belongs to B_t if and only if there is a walk $v_1, \dots, v_t = v$ in G such that $v_i \notin S_i$ for every $1 \leq i < t$. Consequently capture is guaranteed by day t exactly when $B_t \subseteq S_t$.*

Proof. The assertion about walks follows by induction. At $t = 1$ every vertex is an admissible initial position. For the inductive step, $v \in B_{t+1}$ means that some neighbor $u \in B_t$ avoids S_t . Append v to a walk ending at u supplied by the induction hypothesis. Conversely, the penultimate vertex of any such walk belongs to $B_t \setminus S_t$, so its final vertex belongs to B_{t+1} .

A walk ending outside S_t avoids all inspections through day t , whereas every surviving walk is caught that day if its endpoint lies in S_t . This is exactly the stated inclusion. $\square$

Since G has no isolated vertices, a nonempty set has a nonempty open neighborhood. Hence the capture condition is also equivalent to $B_{t+1} = \emptyset$. We use this equivalent form only under that hypothesis. A board with a single room is handled directly: one inspection on the first day suffices. Thus an inability to make a move is never silently treated as capture.

Two monotonicity observations will be useful. Starting from fewer possible positions cannot make a fixed inspection sequence less successful, and adding inspected vertices cannot make it less successful. Both follow by induction from monotonicity of N and set difference in their respective arguments. In particular, probes outside the current possible-position set may be wasted, but they cause no problem for the model or the lower bounds.

2.1 Two initial parity classes

Suppose G is bipartite, with vertex classes V_0, V_1 . Keep track of the two possible initial colors separately:

$$R_1^s = V_s, \quad R_{t+1}^s = N(R_t^s \setminus S_t), \quad s \in \{0, 1\}.$$

Then

$$B_t = R_t^0 \dot{\cup} R_t^1, \quad R_t^s \subseteq V_{s+t-1} \pmod{2}.$$

Indeed, neighborhood and deletion distribute over unions, and every edge reverses color. The two sets therefore stay disjoint while exchanging their current colors each night. Allocations to the two initial classes sum to at most the day's inspection budget. We will state explicitly whether a pair of counts is indexed by initial color or by current color; the latter convention swaps the two counts after every move.

This separates the bookkeeping common to the path, ladder, and grid arguments. The work specific to a graph is to determine how small a neighborhood can be after a given number of inspections, and when a sequence of such bounds can actually be attained.

3 Compressing entire strategies

An isoperimetric inequality bounds one neighborhood. To preserve an optimal search, the replacement sets must also respect containment and remain compatible after successive moves. The following criterion provides that stronger conclusion.

Definition 3.1. A *strategy compression* on G is a map $C : 2^V \rightarrow 2^V$ satisfying

$$A \subseteq B \implies C(A) \subseteq C(B), \tag{6}$$

$$|C(A)| = |A|, \tag{7}$$

$$N(C(A)) \subseteq C(N(A)). \tag{8}$$

Idempotence is an additional property, not part of this definition.

Theorem 3.2 (Strategy comparison and a fixed-state normal form). *Let C be a strategy compression. Every successful strategy from A with prescribed daily budgets $m_1, \dots, m_T$ has a successful strategy from $C(A)$ with the same budgets. If C is idempotent, every starting set fixed by C has an optimal strategy whose possible-position sets and post-inspection survivor sets are all fixed by C .*

Proof. Let A_t be the original possible-position set before day t , and R_t its survivors, so $A_{t+1} = N(R_t)$ and $|A_t| - |R_t| \leq m_t$. Maintain a new actual possible-position set $B_t \subseteq C(A_t)$, initially $B_1 = C(A_1)$. Inspect

$$Q_t = B_t \setminus C(R_t).$$

Monotonicity and equal cardinalities give

$$|Q_t| \leq |C(A_t) \setminus C(R_t)| = |A_t| - |R_t| \leq m_t.$$

The new survivors are $B_t \cap C(R_t)$, and

$$B_{t+1} = N(B_t \cap C(R_t)) \subseteq N(C(R_t)) \subseteq C(N(R_t)) = C(A_{t+1}).$$

When the original survivors are empty, $C(R_t) = \emptyset$, so the new strategy also captures every remaining possibility.

Fixed sets are closed under intersections: if X, Y are fixed, monotonicity gives $C(X \cap Y) \subseteq X \cap Y$, and cardinality forces equality. They are closed under neighborhoods for the same reason: $N(X) \subseteq C(N(X))$ when X is fixed. If C is idempotent, $C(R_t)$ is fixed. Starting the preceding construction from a fixed B_1 , every actual survivor $B_t \cap C(R_t)$ and every next possible-position set is therefore fixed. Restricted strategies are actual strategies, so their optimum equals the unrestricted optimum. $\square$

3.1 Products and simultaneous compression

Lemma 3.3 (Product lift). *A strategy compression on a graph A lifts to $A \square H$ by applying it separately in every H -indexed A -fiber. Idempotence is preserved.*

Proof. Write S_h for the fiber at h . Its neighborhood is

$$N(S)_h = N_A(S_h) \cup \bigcup_{u \sim h} S_u.$$

For the lifted operator $\hat{C}$, the corresponding neighborhood is

$$\begin{aligned} N(\hat{C}(S))_h &= N_A(C(S_h)) \cup \bigcup_{u \sim h} C(S_u) \\ &\subseteq C(N_A(S_h)) \cup \bigcup_{u \sim h} C(S_u) \\ &\subseteq C\left(N_A(S_h) \cup \bigcup_{u \sim h} S_u\right). \end{aligned}$$

The last inclusion uses monotonicity separately on each input subset of the union. Cardinality, monotonicity and idempotence hold fiber by fiber. $\square$

Compositions of strategy compressions are strategy compressions, since

$$N(C(D(S))) \subseteq C(N(D(S))) \subseteq C(D(N(S))).$$

Suppose finitely many such operators strictly decrease a common nonnegative integer potential whenever they change a set. If that potential has a uniform upper bound B , cycling through all operators $B+1$ times reaches a common fixed point for every input. A cycle with a change decreases the potential, and a cycle without a change fixes every operator. The resulting map is a composition with a *uniform* number of factors, hence is monotone and cardinality preserving; it is also idempotent. This avoids assuming that an input-dependent stopping rule preserves monotonicity.

3.2 Odd path fibers

Lemma 3.4 (Odd-path compression). *On P_{2q+1} , $q \geq 1$, number vertices $0, \dots, 2q$. Replace the selected vertices of each parity by the same number of vertices at the beginning of*

$$0, 2, \dots, 2q \quad \text{or} \quad 1, 3, \dots, 2q-1.$$

This is an idempotent, parity-preserving strategy compression.

Proof. Monotonicity, cardinality and idempotence follow from the definition. An even-parity k -set has at least $\min(k, q)$ odd neighbors. If $k \leq q$, omit an unselected even vertex $2j$ and match $2i$ to $2i+1$ for $i < j$, and to $2i-1$ for $i > j$. These matches are distinct and include every selected even vertex. If $k = q+1$, all q odd vertices are neighbors. A nonempty odd-parity k -set has at least $k+1$ even neighbors: its consecutive blocks in the spacing-two order each have one more neighbor than their size, and different blocks have disjoint neighbor intervals. The two prefixes attain these bounds, and their neighborhoods are prefixes of the opposite parity. Consequently each part of $N(C(S))$ is contained in the corresponding same-cardinality prefix of $N(S)$, proving (8) even when S contains both colors. $\square$

For several odd coordinates, combine their lifted operators. Every changing operation decreases the sum of coordinates, so simultaneous compression has the normal form of Theorem 3.2. Its fixed sets satisfy $v \in S \implies v - 2e_i \in S$ whenever the latter vertex belongs to the box. These are lower ideals separately in each coordinate-parity pattern. They need not be lower ideals in the whole checkerboard color: for example, $\{(1, 1)\}$ in a 3×3 square is fixed by the two path operators but does not contain $(0, 0)$.

3.3 Exact recurrences on odd cylinders

Let $G = H \square P_{2q+1}$, where H is any finite graph and $q \geq 1$. The product lift and Theorem 3.2 give an exact normal form from the full board. It uses two vectors indexed by $V(H)$: a_v^0 counts even path coordinates in fiber v , and a_v^1 counts odd path coordinates. Their ranges are

$$0 \leq a_v^0 \leq q + 1, \quad 0 \leq a_v^1 \leq q.$$

Thus they refer to the parity of the *path coordinate*, and do not require H to be bipartite. Define

$$\phi_0(k) = \min(k, q), \quad \phi_1(k) = \begin{cases} 0, & k = 0, \\ k + 1, & k > 0. \end{cases}$$

For survivor vectors $b^0 \leq a^0$, $b^1 \leq a^1$, the exact next state $F(b)$ is

$$F(b)_v^0 = \max \left(\phi_1(b_v^1), \max_{u \sim v} b_u^0 \right), \quad (9)$$

$$F(b)_v^1 = \max \left(\phi_0(b_v^0), \max_{u \sim v} b_u^1 \right), \quad (10)$$

with an empty maximum interpreted as zero. Path edges supply the first term, and H -edges the second. All contributions in a fiber are prefixes of the same parity, so their union has exactly the maximum length. The required number of inspections is

$$d(a, b) = \sum_{v \in V(H)} (a_v^0 - b_v^0 + a_v^1 - b_v^1).$$

They are realized by inspecting the removed suffix of each prefix.

For a fixed budget m , form an edge $a \rightarrow F(b)$ whenever $b \leq a$ and $d(a, b) \leq m$. Then $T_m(a)$ is exactly the shortest-path distance to zero in this state graph, or infinity if zero is unreachable. Equivalently,

$$T_m(a) = 1 + \min_{\substack{b \leq a \\ d(a, b) \leq m}} T_m(F(b)), \quad T_m(0) = 0, \quad (11)$$

where the shortest-path interpretation specifies the solution of the possibly cyclic equations. For all budgets simultaneously, let $W_t(a)$ be the least constant budget that wins within t days. Then

$$W_{t+1}(a) = \min_{b \leq a} \max \{d(a, b), W_t(F(b))\}, \quad (12)$$

with $W_0(0) = 0$, $W_0(a) = \infty$ for $a \neq 0$. Backpointers in either recurrence give actual inspection sets.

If $A = |V(H)|$, the number of states is exactly

$$K = ((q + 1)(q + 2))^A \leq (2q + 2)^{2A}.$$

There are at most K^2 state-survivor pairs. Explicit construction of their transitions takes $O((A + |E(H)|)K^2)$ elementary operations, apart from integer bit costs. For fixed H , this is polynomial in the odd longitudinal length. Any successful path can have cycles removed, so a finite optimum is at most $K - 1$. This proves an exact algorithm, not a uniform polynomial bound in dimension or a closed time formula.

When H is bipartite, one may instead index vectors by global color. For a coloring χ , put $\varepsilon_p(v) = p - \chi(v) \pmod{2}$. A single current color has transition

$$[\Phi_p(b)]_v = \max \left(\phi_{\varepsilon_p(v)}(b_v), \max_{u \sim v} b_u \right).$$

The two current-color vectors then become $(\Phi_1(b^1), \Phi_0(b^0))$. This convention is useful when comparing the cylinder recurrence with the two-color grid arguments below.

3.4 Square fibers and equal-sided boxes

Abramovskaya, Fomin, Golovach and Pilipczuk [1, Section 3.1, Lemma 2] prove that diagonal shifts do not increase the neighborhood size of a one-color set in a square. The following fiber calculation strengthens the scalar comparison to (8) and applies it to both colors. That stronger statement is what allows whole strategies to be compressed.

Lemma 3.5 (Square diagonal compression). *On $\{0, \dots, n-1\}^2$, compress each line $x + y = k$ toward increasing priority for smaller y , keeping its selected cardinality. This is an idempotent, parity-preserving strategy compression. The same is true on lines $x - y = k$, again toward smaller y .*

Proof. Only the neighborhood inclusion needs proof. Consecutive square diagonals have lengths differing by one. Index both from the common increasing- y end. From a diagonal of length L to one of length $L + 1$, a selected index j has neighbors $j, j + 1$. A nonempty a -set therefore has at least $a + 1$ neighbors; a prefix attains this bound with a prefix. From length L to length $L - 1$, the indices are $j - 1, j$, clipped to the valid range. A proper nonempty a -set has at least a neighbors, and the full set has $L - 1$. Indeed, if it has c consecutive blocks and occupies e endpoints, its neighbor count is $a + c - e$; for a proper set $c \geq e$. Prefixes again attain the bounds with prefixes.

Fix an output diagonal. Its two input diagonals, after compression, contribute prefixes from the same end. Their union is the longer prefix. Each contributing length is no larger than its original contribution, hence no larger than the original union's size. The compressed union is therefore contained in the prefix selected by compressing that original neighborhood. This argument works on either parity and on their union. Reflecting x proves the other diagonal direction. $\square$

Theorem 3.6 (Normal form on equal-sided boxes). *On P_n^d , $n \geq 2$, an optimal strategy from the full board can be chosen so that its possible-position sets and survivors are lower ideals for the legal moves*

$$v \mapsto v + e_i - e_j, \quad v \mapsto v - e_i - e_j, \quad i < j.$$

The conclusion respects arbitrary prescribed daily budgets. The same operators are available within every group of equal-length coordinates in an arbitrary Cartesian box. Odd-coordinate path compressions can be included simultaneously.

Proof. Lift both square operators to every pair $i < j$ of equal coordinates. They compress toward smaller x_j on lines $x_i + x_j = k$ and $x_i - x_j = k$. Use the potential

$$\mathcal{E}(S) = \sum_{v \in S} \sum_{i=1}^d i v_i.$$

A nontrivial plus-diagonal transfer by $\delta > 0$ decreases it by $(j - i)\delta$, and a minus-diagonal transfer by $(i + j)\delta$. An odd-coordinate prefix transfer also strictly decreases it. The potential is uniformly bounded; on P_n^d the bound $n^d(n - 1)d(d + 1)/2$ suffices. Uniform cycling therefore gives an idempotent strategy compression. Its fixed sets are precisely the ideals for the displayed local moves, with the additional $-2e_i$ conditions when odd-coordinate operators are included. Apply Theorem 3.2. $\square$

In dimension two the two-diagonal fixed sets are the downward pyramidal sets of the square isoperimetric argument. They differ from ordinary checkerboard corner ideals. In any dimension they give an exact finite recurrence over their fixed family, since both neighborhoods and intersections remain fixed. The theorem does not bound the number of these ideals by a polynomial in d or in n .

Remark 3.7 (Why even path factors require another idea). Every parity-preserving strategy compression on an even path is the identity. Each endpoint is the unique degree-one vertex in its color. Applying (8) to its singleton forces that singleton to be fixed. Fixed sets are closed under neighborhoods, so every N^t of either endpoint is fixed. These are the spacing-two prefixes from the two ends. Intersections of suitable prefixes isolate every singleton; fixed intersections follow from monotonicity and cardinality, without idempotence. Thus all singletons, and consequently all sets, are fixed. This rules out a nontrivial even-path factor reduction within this specific comparison framework. It does not rule out richer state labels or different comparison theorems.

4 Exact neighborhood profiles on odd rectangles

Let $G = P_{2a+1} \square P_{2b+1}$, where $a \geq b \geq 1$, with coordinates (x, y) ; thus x is the long coordinate. Write V_0, V_1 for its even and odd checkerboard classes, and

$$E = |V_0| = \frac{|G| + 1}{2}, \quad M = |V_1| = E - 1.$$

Define the two open-neighborhood profiles by

$$g_p(k) = \min\{|N(S)| : S \subseteq V_p, |S| = k\}.$$

It is convenient to use the integer functions

$$R(k) = \lceil \sqrt{k} \rceil, \quad Q(k) = \min\{r \geq 1 : k \leq r(r - 1)\}.$$

In particular $R(0) = 0$ and $Q(0) = 1$.

Theorem 4.1 (Odd-rectangle profiles and compatible orders). *Both profiles have $g_p(0) = 0$. For nonempty sets,*

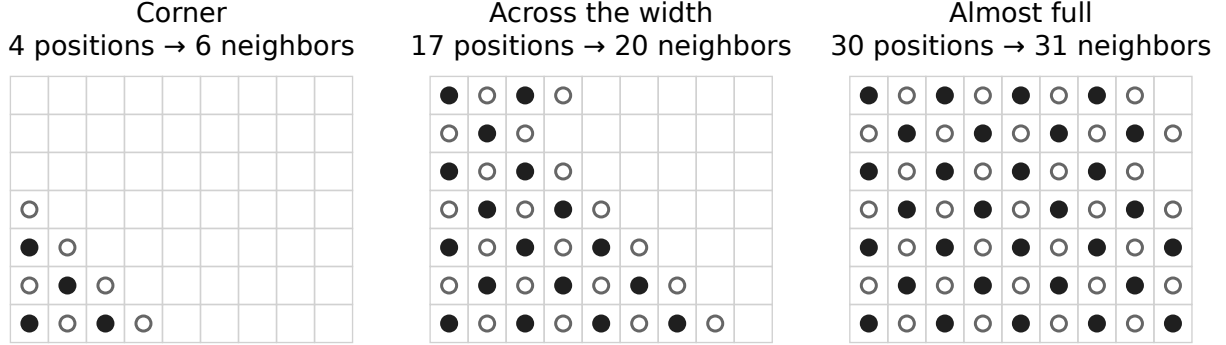
$$g_0(k) = k + \min\{R(k), b, R(E - k) - 1\}, \quad 1 \leq k \leq E, \quad (13)$$

$$g_1(k) = k + \min\{Q(k), b + 1, Q(M - k)\}, \quad 1 \leq k \leq M. \quad (14)$$

Order each parity by increasing $(x + y, x)$, and let $I_p(k)$ be its prefix of size k . Every prefix attains the corresponding minimum, and

$$N(I_p(k)) = I_{1-p}(g_p(k)).$$

The order fills the short direction first within a diagonal. Its direction is significant on an unequal rectangle. For example, the opposite tie order on 7×5 gives the five-element even prefix $\{(0, 0), (0, 2), (1, 1), (2, 0), (4, 0)\}$, with eight neighbors; the set $\{(0, 0), (0, 2), (0, 4), (1, 1), (1, 3)\}$ has seven.



Filled circles: source set. Open circles: its possible neighbors.

Figure 2: Minimizing even prefixes on a 9×7 rectangle. The three regimes are controlled by a corner, the short width, and the complement.

4.1 Quadrant bounds

Apply the odd-coordinate compressions of Lemma 3.4 in both directions. Their common fixed sets are lower closed under decreasing either coordinate by two, and compression does not increase the neighborhood size. We may therefore assume that each coordinate-parity pattern of S is a lower Ferrers diagram. Let e_i and o_i be the selected counts in rows $y = 2i$ and $y = 2i + 1$, respectively. These are separately nonincreasing sequences. For source color 0 they count even- x and odd- x vertices, whereas for source color 1 they count odd- x and even- x vertices.

Lemma 4.2 (Quadrant area and row bounds). *Regard a finite set with this Ferrers property as a subset of the infinite nonnegative quadrant. Put $\delta = |N(S)| - |S|$. If S has color 0, then*

$$e_i > 0 \implies e_i + i \leq \delta, \quad o_i > 0 \implies o_i + i + 1 \leq \delta, \quad |S| \leq \delta^2.$$

If S has color 1, then

$$e_i > 0 \implies e_i + i + 1 \leq \delta, \quad o_i > 0 \implies o_i + i + 1 \leq \delta, \quad |S| \leq \delta(\delta - 1).$$

A nonempty set has surplus at least one in color 0 and at least two in color 1.

Proof. The sequences eventually vanish. A nonempty odd- x prefix of length t has $t + 1$ even- x neighbors in the quadrant; an even- x prefix has t odd- x neighbors. Contributions from neighboring rows are also prefixes, so their unions are given by maxima. Subtracting the source row sizes therefore yields, in color 0,

$$\delta = (o_0 - e_0)_+ + \sum_{i \geq 0} \left[\max\{e_i - o_i, \mathbf{1}_{\{o_i > 0\}}\} + (o_i - e_{i+1})_+ \right]. \quad (15)$$

For color 1 the identity is

$$\delta = \max\{\mathbf{1}_{\{e_0 > 0\}}, o_0 - e_0\} + \sum_{i \geq 0} \left[(e_i - o_i)_+ + \max\{o_i - e_{i+1}, \mathbf{1}_{\{e_{i+1} > 0\}}\} \right]. \quad (16)$$

All terms displayed are nonnegative.

Suppose first that S has color 0 and $e_i > 0$. In (15), each of the first i paired summands is at least one. If its odd count is positive, the first term is at least one; otherwise its even count is positive. The tail from i onward is at least the telescoping sum

$$\sum_{j \geq i} ((e_j - o_j) + (o_j - e_{j+1})) = e_i.$$

Thus $\delta \geq i + e_i$. If $o_i > 0$, every preceding pair contributes at least one, as does the first term of pair i . The second term of pair i and the subsequent pairs have sum at least o_i , by the same telescoping argument. This gives $\delta \geq i + 1 + o_i$.

For color 1 and $e_i > 0$, the initial term of (16) is at least one, and the first i pairs each contribute at least one because $e_{j+1} > 0$ for $j < i$. The remaining tail is at least e_i . If $o_i > 0$, the initial term is again at least one. Every preceding pair contributes at least one: either $e_{j+1} > 0$, or the positive o_j appears in its second term. The tail after the first term of pair i is at least o_i . These are the two claimed bounds for color 1.

Finally sum the row bounds. In color 0 the maximum possible total is

$$\sum_{i \geq 0} \max(\delta - i, 0) + \sum_{i \geq 0} \max(\delta - i - 1, 0) = \delta^2.$$

In color 1 both sums have the second form, giving $\delta(\delta - 1)$. The row inequalities also give the stated positive surplus bounds for nonempty sets. $\square$

4.2 The two far edges

Lemma 4.3 (Corner, strip, or complement). *Let $S \subseteq V_p$ have the Ferrers property. Put $\delta = |N_G(S)| - |S|$, and let F and Z count its vertices on the far edges $x = 2a$ and $y = 2b$, respectively. If exactly one of F, Z is positive, then $\delta \geq b + p$. If neither is positive, the quadrant bounds apply to S without change. If both are positive, $N_G(S)$ covers both near edges in color $1 - p$, and the set*

$$V_{1-p} \setminus N_G(S),$$

reflected in both coordinate midlines, has the Ferrers property and no far-edge vertices.

Proof. Viewing the same S in the quadrant adds exactly one neighbor beyond the rectangle for every selected far-edge vertex. These $F + Z$ vertices are distinct, even when the far corner is selected. Hence

$$\delta_{\text{quadrant}} = \delta + F + Z. \quad (17)$$

If $F = 0$ and $Z > 0$, the last even row has $e_b = Z$. The row bound in Lemma 4.2 gives $\delta + Z \geq Z + b + p$. If $F > 0$ and $Z = 0$, swap the axes to get the stronger bound $\delta \geq a + p$. If $F = Z = 0$, there is no clipping.

Suppose now that both are positive. In color 0, the Ferrers property forces every even- x vertex of row $y = 0$ and every even- y vertex of column $x = 0$ into S . They cover both opposite-color near edges. In color 1, far- x occupancy forces the full even- x row $y = 1$, covering the even- x near edge $y = 0$. Far- y occupancy forces $(1, 2i) \in S$ for every $0 \leq i \leq b$, covering the even- y near edge $x = 0$.

The neighborhood of a fixed set is fixed under the two path operators, so its complement is upper closed within each coordinate-parity pattern. The two reflections convert it to a lower Ferrers set. They preserve checkerboard color because both side lengths are odd. The near-edge coverage just proved means the reflected complement has no far-edge vertices, as required. $\square$

4.3 Lower bounds in Theorem 4.1

Let $S \subseteq V_0$, $|S| = k > 0$, and put $\delta = |N(S)| - k$. If $\delta \geq b$ the desired lower bound is immediate. Otherwise Lemma 4.3 leaves two possibilities. With no clipping, $k \leq \delta^2$. With both far edges occupied, let T be the reflected opposite-color complement of $N(S)$. It has size $t = M - k - \delta$. Before reflection, its neighborhood is contained in $V_0 \setminus S$, so its quadrant surplus is at most $\delta + 1$. For $\delta \geq 1$, Lemma 4.2 gives

$$M - k - \delta = t \leq \delta(\delta + 1), \quad k \geq E - (\delta + 1)^2.$$

If $\delta = -1$ or 0 , T must be empty, since a nonempty quadrant color-1 set has surplus at least two. This yields respectively $k = E$ or $k = M$, the same conclusion. No smaller δ is possible: pairing along each row, then vertically in the unpaired last column, gives a matching covering all vertices except the far even corner, and hence $|N(S)| \geq k - 1$. In all cases,

$$\delta \geq \min\{R(k), b, R(E - k) - 1\}.$$

For $S \subseteq V_1$, the same matching gives $\delta \geq 0$. Below $b + 1$, the unclipped case gives $k \leq \delta(\delta - 1)$. In the complement case, T has size $E - k - \delta$ and quadrant color-0 surplus at most $\delta - 1$. The case $\delta = 0$ is impossible: T is nonempty because $k \leq M < E$, but would have negative surplus. For $\delta \geq 1$,

$$E - k - \delta \leq (\delta - 1)^2, \quad k \geq M - \delta(\delta - 1).$$

The unclipped nonempty case also has $\delta \geq 1$. Thus

$$\delta \geq \min\{Q(k), b + 1, Q(M - k)\}.$$

These prove the lower bounds in both formulas.

For comparison, the two exact profiles of any bipartite graph obey the complement-inverse identity

$$g_1(y) = E - \max\{x : g_0(x) \leq M - y\}. \quad (18)$$

Indeed, the existence of nonadjacent sets of sizes x and y in the two colors is equivalent to $g_0(x) \leq M - y$, and also to $g_1(y) \leq E - x$. Thus the second formula can alternatively be checked from the first by exact inversion.

4.4 Attainment and neighborhood nesting

The vertices on diagonal $x + y = d$ have x in the interval

$$L_d = \max(0, d - 2b) \leq x \leq \min(2a, d),$$

whose length is

$$\ell(d) = \min(2a, d) - \max(0, d - 2b) + 1, \quad 0 \leq d \leq 2a + 2b.$$

A nonempty prefix ends after j vertices of diagonal d , for some $1 \leq j \leq \ell(d)$, and has size

$$k = \sum_{\substack{s < d \\ s \equiv d \pmod{2}}} \ell(s) + j. \quad (19)$$

All earlier opposite-color diagonals belong to its neighborhood. On diagonal $d + 1$, each selected x contributes x or $x + 1$ when the corresponding step stays inside the board. Their union is the initial interval there, of length

$$f(d, j) = j + 1 - \mathbf{1}_{\{d \geq 2b\}} - \mathbf{1}_{\{L_{d+j-1}=2a\}}. \quad (20)$$

The first indicator removes the unavailable step past the short-direction boundary, and the second removes the step past the long-direction boundary. At the far corner both are present and $f = 0$. The initial minority diagonal $d = 1$ additionally covers the corner on diagonal zero, already included among the earlier opposite diagonals. Consequently

$$|N(I_p(k))| = \sum_{\substack{s < d \\ s \not\equiv d \pmod{2}}} \ell(s) + f(d, j),$$

and this neighborhood is exactly a prefix of the other color.

To evaluate its surplus, let $\Delta(d)$ be the earlier opposite-color sum minus the earlier same-color sum. The recurrence $\Delta(0) = 0$, $\Delta(d + 1) = \ell(d) - \Delta(d)$ gives

$$\Delta(d) = \begin{cases} \lceil d/2 \rceil, & d \leq 2b, \\ b + (d \bmod 2), & 2b \leq d \leq 2a, \\ a + b - \lfloor d/2 \rfloor + (d \bmod 2), & d \geq 2a. \end{cases}$$

The formulas agree at their shared endpoints. The exact surplus is $\Delta(d) + 1$ minus the two indicators in (20). Complete even diagonals through sum $2r - 2$ contain r^2 vertices, and complete odd diagonals through sum $2r - 1$ contain $r(r + 1)$. Substitution into (19) yields

Color	Size range	Prefix surplus
0	$1 \leq k \leq b^2$	$R(k)$
0	$b^2 < k < E - b^2$	b
0	$E - b^2 \leq k \leq E$	$R(E - k) - 1$
1	$1 \leq k \leq b(b + 1)$	$Q(k)$
1	$b(b + 1) < k < M - b(b + 1)$	$b + 1$
1	$M - b(b + 1) \leq k \leq M$	$Q(M - k)$

For a square, the shared minority endpoint has surplus $b + 1$ in both rows; empty ranges are omitted. The table equals the two lower bounds. It proves attainment and the compatible-neighborhood assertion, completing the proof of Theorem 4.1.

4.5 An exact two-count game

Corollary 4.4. *For every daily budget on every odd-by-odd rectangle, the unrestricted minimum capture time is exactly the shortest-path distance from (E, M) to $(0, 0)$ in the following two-count state graph:*

$$(u, v) \longrightarrow (g_1(j), g_0(i)), \quad 0 \leq i \leq u, \quad 0 \leq j \leq v, \quad u + v - i - j \leq m.$$

A shortest path gives an optimal physical strategy by deleting suffixes of the two current-color prefix orders. The reduction also respects arbitrary prescribed daily budgets.

Proof. Define

$$C(S) = I_0(|S \cap V_0|) \cup I_1(|S \cap V_1|).$$

It is monotone, cardinality preserving and idempotent. By the exact profiles and nesting,

$$N(C(S)) \cap V_{1-p} = I_{1-p}(g_p(|S \cap V_p|)) \subseteq I_{1-p}(|N(S \cap V_p)|).$$

The last set is the corresponding part of $C(N(S))$. Apply Theorem 3.2. Every displayed transition is also the exact neighborhood of the indicated survivors, so it has a physical realization. Shortest paths, with infinity for an unreachable zero state, settle both feasibility and minimum time. $\square$

The corollary is an exact recurrence for every parameter, rather than a closed expression for its shortest-path distance. It avoids imposing an unproved sequential allocation to the two initial colors: both may receive inspections on any day.

5 An exact isoperimetric order in every all-odd box

The preceding rectangle theorem extends to every dimension. The extension uses a weighted exchange argument: compression in lower-dimensional faces first makes neighborhood size a sum of local costs, and an order comparison then identifies exchanges that cannot increase that sum. Parity-restricted isoperimetry on hypercubes was studied by Körner and Wei [11], and local-to-global methods for Cartesian-product vertex boundaries by Bezrukov and Serra [4]. Here the neighborhood is the open neighborhood of a single color class. We give the face compressions and their exchange argument explicitly, rather than infer the result from a theorem about closed vertex boundaries.

Delete any factors of side length one, and write the remaining box as

$$B = \prod_{i=1}^d \{0, \dots, L_i\}, \quad 2 \leq L_1 \leq \dots \leq L_d, \quad L_i \text{ even.}$$

Thus coordinates are ordered from the shortest side to the longest. Let B_p be its color- p class, where the color of x is the parity of $|x| = \sum_i x_i$. Within either color, order vertices by increasing $|x|$ and then by *decreasing* lexicographic coordinates. Write $x \prec y$ for this order and $I_p(k)$ for its first k vertices. The direction of the lexicographic tie is part of the definition.

Theorem 5.1 (Compatible isoperimetric nesting in all-odd boxes). *For every $S \subseteq B_p$,*

$$|N(S)| \geq |N(I_p(|S|))|.$$

Moreover, $N(I_p(k))$ is a prefix in the opposite color. Consequently the minimum capture time for every inspection budget on B is given by an exact recurrence on $O(|B|^2)$ states, and optimal inspection sets can be recovered from that recurrence.

For $d = 1$ the minimizing order is the spacing-two prefix order of Lemma 3.4. For $d = 2$, putting the shorter coordinate first and ordering it decreasingly within a weight layer is equivalent to the increasing-long-coordinate convention of Theorem 4.1. These provide the induction bases. In particular, the higher-dimensional assertion does not assume a general Cartesian-product isoperimetric principle.

5.1 Prefix neighborhoods and a local cost identity

For a nonzero vertex z , let $j(z)$ be its last positive coordinate and put

$$\ell(z) = z - e_{j(z)}.$$

This is its earliest lower neighbor in decreasing lexicographic order.

Lemma 5.2 (Nesting of prefix neighborhoods). *For the stated weight and lexicographic order, the neighborhood of a prefix is a prefix of the opposite color. This statement holds even without the assumptions that the side lengths are odd and sorted.*

Proof. Within a fixed weight layer, ℓ preserves order, allowing ties. Indeed, suppose $z \prec w$ first differ in coordinate i , so $z_i > w_i$. Equal weights imply that w has a positive coordinate after i . If z does too, neither predecessor operation changes the first difference. Otherwise $j(z) = i$ and $z_i - 1 \geq w_i$. A strict inequality preserves the order. In the equality case the tail of w has total weight one; removing its unique unit gives $\ell(z) = \ell(w)$.

Consider a nonempty prefix whose last, possibly partial, layer has weight r . Every smaller opposite-color layer is completely covered: its nonzero vertices have lower neighbors in complete source layers. When weight zero belongs to the opposite color, it is covered by the first odd layer. No vertex above layer $r + 1$ is covered. On layer $r + 1$, a vertex is covered precisely when its earliest lower neighbor belongs to the selected prefix in layer r . The order preservation just proved makes these covered vertices an initial interval of layer $r + 1$. The empty prefix is immediate. $\square$

Call a set *pair-compressed* if, after fixing all but any two coordinates, its selected vertices in either free-coordinate parity form a prefix of the inherited order. Define

$$a(0) = d, \quad a(x) = d - j(x) + 1 - \mathbf{1}_{\{x_{j(x)} = L_{j(x)}\}} \quad (x \neq 0). \quad (21)$$

Lemma 5.3 (Neighborhood cost of a pair-compressed set). *If $d \geq 2$ and $\emptyset \neq S \subseteq B_p$ is pair-compressed, then*

$$|N(S)| = \sum_{x \in S} a(x) + \mathbf{1}_{\{p=1\}}. \quad (22)$$

Proof. For every nonzero z , we claim

$$z \in N(S) \iff \ell(z) \in S.$$

Only the forward implication needs proof. Choose a neighbor $x \in S$ of z . The vertices x and $\ell(z)$ differ in at most two coordinates. If x is a lower neighbor, then $\ell(z)$ is no later than x in their common weight layer. If x is an upper neighbor, $\ell(z)$ has weight two less than x . Pair compression therefore puts $\ell(z)$ in S . If only one coordinate differs, any second coordinate can be used; one exists because $d \geq 2$.

For $x \neq 0$ with last positive coordinate J , the preimages of x under ℓ are obtained by adding one in coordinate J , if it is not full, or by adding one in any coordinate after J . Their number is $a(x)$. The origin has the d unit vectors as preimages. Thus the sum in (22) counts every nonzero neighbor exactly once.

If $p = 1$, a nonempty pair-compressed set contains a unit vector. Starting with any of its vertices of weight greater than one, decrease by two in a coordinate that is at least two, or decrease by one in two positive coordinates. Each move stays in a pair fiber and goes earlier in its order, so it preserves membership. The process ends at weight one. The origin is therefore an additional neighbor when $p = 1$, and it can never be a neighbor when $p = 0$. $\square$

5.2 Face compression and its partial order

Suppose $d \geq 3$ and the isoperimetric theorem has been proved in dimension $d - 1$. On each fixed color define a partial order $\leq_P$ as the reflexive transitive closure of

$$x <_P y \quad \text{if } x \prec y \text{ and } x_i = y_i \text{ for at least one } i. \quad (23)$$

All generating relations increase the total order, so this is indeed a partial order. Its ideals are exactly the sets whose restriction to every codimension-one fiber is a prefix.

The inductive isoperimetric theorem and compatible neighborhood nesting make the prefix map on each $(d - 1)$ -dimensional box a strategy compression: it is monotone, preserves cardinality, and sends each color's neighborhood into the corresponding prefix of the original neighborhood's size. The remaining side lengths are still sorted, and their order is the restriction of the global order to a fixed-coordinate fiber. By Lemma 3.3, compressing all such fibers for one coordinate index is a strategy compression of B .

Cycle through the d indices. Every change strictly decreases the sum of the selected vertices' positions in the global weight and lexicographic order. This nonnegative integer is bounded by $|B|(|B| - 1)/2$. The simultaneous-compression argument in Section 3 therefore gives a uniform finite composition whose output is fixed by every face compression. In particular, every $S \subseteq B_p$ can be replaced by a P -ideal of the same size without increasing its neighborhood.

Every P -ideal is pair-compressed. A two-coordinate fiber lies inside a codimension-one fiber obtained by fixing a coordinate outside the pair; such a coordinate exists because $d \geq 3$. The restriction of a prefix to a subfiber is a prefix of its inherited order. Thus Lemma 5.3 applies to the resulting ideals.

5.3 The exchange comparison

The following elementary comparison is the step that links the local cost identity to the global order.

Lemma 5.4 (Terminal-coordinate comparison). *Let $d \geq 3$. If same-parity vertices satisfy $x \prec y$ and $x_d = 0 < y_d$, then $x \leq_P y$. Also, if $x \prec y$ and $x_d < y_d = L_d$, then $x \leq_P y$.*

Proof. First, if $u \leq v$ coordinatewise and their weights have the same parity, then $u \leq_P v$. Perform all required increments of size two within individual coordinates, and then pair the remaining increments of size one. Each step increases weight by two and changes at most two coordinates. Since $d \geq 3$, it leaves a coordinate unchanged and is a generator of (23).

For the first assertion, a coordinate agreement between x and y already gives a generator. Assume henceforth that all coordinates differ. If $|x| < |y|$ and some $i < d$ has $x_i > y_i$, transfer $t = x_i - y_i$ units from coordinate i to coordinate d of x , obtaining z . This is legal because $0 < t \leq L_i \leq L_d$ and $x_d = 0$. We have $x \prec z$ within their common weight layer, while $z \prec y$ follows from $|z| = |x| < |y|$. The vertex z retains $d - 2 \geq 1$ coordinates of x and agrees with y in coordinate i . Hence $x \prec_P z \prec_P y$. If there is no such excess coordinate, then $x \leq y$ coordinatewise, and the preceding observation applies.

It remains to consider $|x| = |y|$. Since all coordinates differ, $x_1 > y_1$. If a middle coordinate $2 \leq i \leq d - 1$ has $x_i > y_i$, use the same transfer to coordinate d . Again $x \prec z$, and $z \prec y$ because $z_1 = x_1 > y_1$. The same coordinate agreements give the two generators. Otherwise all middle coordinates increase: $x_i < y_i$ for $2 \leq i \leq d - 1$. Weight equality gives

$$x_1 - y_1 = y_d + \sum_{i=2}^{d-1} (y_i - x_i).$$

Transfer y_d units from the first to the last coordinate of x . The resulting vertex z is valid, since

$$z_1 = y_1 + \sum_{i=2}^{d-1} (y_i - x_i) > y_1 \geq 0, \quad z_d = y_d \leq L_d.$$

It satisfies $x \prec z \prec y$, shares the middle coordinates with x , and shares the last coordinate with y . This proves the first assertion.

The coordinate complement $\rho(x)_i = L_i - x_i$ reverses the weight and lexicographic order, and it preserves coordinate agreements. It therefore reverses the generated partial order. Applying the first assertion to $\rho(y) \prec \rho(x)$ proves the second assertion. $\square$

Lemma 5.5 (Cost comparison). *If $x \prec y$ have the same parity and $a(x) > a(y)$, then $x \leq_P y$.*

Proof. If $x_d = y_d$, they share a coordinate. If $x_d = 0 < y_d$, use the first part of Lemma 5.4. The case $y_d = 0 < x_d$ cannot give a strict cost decrease, since $a(x) \leq 1$ and $a(y) \geq 1$. Finally, if both last coordinates are positive, both costs are zero or one. A strict decrease forces $x_d < L_d = y_d$, so the second part of that lemma applies. $\square$

Proof of Theorem 5.1. The path and rectangle bases were noted above. In dimension $d \geq 3$, the inductive face compressions replace an arbitrary $S \subseteq B_p$ by a P -ideal of the same size and with no larger neighborhood. The empty set needs no further argument.

If a nonempty P -ideal is not a global prefix, let x be its earliest missing vertex and y its latest included vertex. Then $x \prec y$. They are incomparable in P : $x \leq_P y$ would contradict ideality, and $y \leq_P x$ would contradict the global order. By Lemma 5.5, $a(x) \leq a(y)$.

Replace y by x . Removing y preserves ideality because every proper successor of y is later in the global order and hence absent. Adding x preserves ideality because every proper predecessor of x is earlier and hence present; y is not such a predecessor. The new set has the same positive cardinality and color. The neighborhood cost identity (22) shows that its neighborhood does not increase. The sum of global positions strictly decreases, so iteration ends at the global prefix of that size. This proves the isoperimetric inequality. Lemma 5.2 gives compatible nesting, completing the induction. $\square$

5.4 Profiles, exact capture times, and optimal inspections

Write $V = |B|$, $E = (V + 1)/2$, and $M = (V - 1)/2$. The theorem supplies the complete profiles by a direct cumulative-sum construction:

$$g_p(0) = 0, \quad g_p(k) = \sum_{x \in I_p(k)} a(x) + \mathbf{1}_{\{p=1\}} \quad (k > 0). \quad (24)$$

For $d = 1$ these expressions also agree with the odd-path profiles. Enumerate the vertices, order each color as specified, evaluate (21), and take cumulative sums. Using a mixed-radix lexicographic index, sorting requires $O(V \log V)$ integer comparisons; computing weights, indices and costs directly requires $O(dV)$ additional operations. This is a constructive profile algorithm, without an isoperimetric optimization subroutine.

Corollary 5.6 (Feasibility by one profile maximum). *For every nontrivial all-odd box, the minimum feasible daily budget is*

$$h(B) = 1 + \max_{0 \leq k \leq E} (g_0(k) - k) = 1 + \max_{0 \leq k \leq E} \sum_{x \in I_0(k)} (a(x) - 1). \quad (25)$$

The one-room box has threshold one by direct inspection.

Proof. We verify the hypotheses of the general hunter-number criterion of Bolkema and Groothuis [5, Theorem 16]. That theorem states that a bipartite graph with compatible isoperimetric nesting has hunter number $1 + \min(u_0, u_1)$ when the maximum neighborhood surpluses $u_p = \max_k (g_p(k) - k)$ differ by at most one.

For completeness, the exact profile duality gives the required relation between these maxima. Define

$$J(z) = \max\{x : 0 \leq x \leq E, g_0(x) \leq z\}, \quad 0 \leq z \leq M.$$

Then

$$g_1(y) = E - J(M - y). \quad (26)$$

Indeed, a majority set of size x with at most $M - y$ neighbors leaves a minority y -set with no edges to it, giving $g_1(y) \leq E - x$. Conversely, the majority complement of the neighborhood of a minimizing minority y -set has size $E - g_1(y)$ and at most $M - y$ neighbors. These two choices prove (26). Since $E = M + 1$, it follows that

$$u_1 = 1 + \max_{0 \leq z \leq M} (z - J(z)).$$

This last maximum equals u_0 . If $J(z) < E$, let $x = J(z) + 1$. Then $g_0(x) > z$, so $z - J(z) \leq g_0(x) - x \leq u_0$. If $J(z) = E$, the absence of isolated vertices implies $g_0(E) = M$; thus $z = M$ and $z - J(z) = -1 \leq u_0$. For the reverse inequality choose a positive x attaining u_0 . Such an x exists even when $u_0 = 0$, because $g_0(1) \geq 1$ then forces $g_0(1) = 1$. For $z = g_0(x) - 1$ we have $0 \leq z < M$ and $J(z) \leq x - 1$, hence $z - J(z) \geq g_0(x) - x = u_0$. Therefore $u_1 = u_0 + 1$.

Theorem 5.1 supplies compatible nesting, so the cited criterion yields $h(B) = 1 + u_0$. Its general lower-bound theorem is being used here; profile duality alone is not a lower-bound proof. The cumulative-sum formula follows from (24). $\square$

The upper bound also admits a direct description. Spend all $m = u_0 + 1$ inspections on one current prefix cohort. If it is not yet captured, a majority step decreases its size by at least one,

while a minority step does not increase it, because $u_1 = u_0 + 1$. Thus it clears after finitely many steps. Clear the other initial cohort next. This proves feasibility without asserting that serial allocation minimizes the time.

For example, the partial-sum scan gives

B	3×3	5×5	$3 \times 3 \times 3$	$3 \times 5 \times 5$	$5 \times 5 \times 5$	$3 \times 3 \times 3 \times 3$
$h(B)$	2	3	5	8	12	12

Feasibility therefore needs only the profile scan. Minimum capture time uses the following two-count recurrence.

Define

$$C(S) = I_0(|S \cap B_0|) \cup I_1(|S \cap B_1|).$$

Isoperimetry and nesting imply that C is an idempotent strategy compression. Theorem 3.2 therefore gives an optimal strategy from the full board in which both current-color parts and both survivor parts are prefixes.

A state is the pair (u, v) of counts in the *current* colors. Choosing survivor counts $0 \leq i \leq u$ and $0 \leq j \leq v$ costs $u + v - i - j$ inspections and gives the exact next state

$$(u, v) \longrightarrow (g_1(j), g_0(i)). \quad (27)$$

The inspection sets are the suffixes $I_0(u) \setminus I_0(i)$ and $I_1(v) \setminus I_1(j)$. Thus every transition is physically realized, and every unrestricted successful search is matched by a path in this state graph.

For a fixed daily budget m , the optimum is the shortest-path distance from (E, M) to $(0, 0)$, with value infinity when there is no such path. There are $(E + 1)(M + 1) = O(V^2)$ states. If $u + v \leq m$, one final day captures every possibility. Otherwise, additional inspections cannot hurt, so it suffices to allocate exactly m useful inspections. For

$$\max(0, m - v) \leq p \leq \min(m, u),$$

the corresponding transition is

$$(u, v) \longrightarrow (g_1(v - m + p), g_0(u - p)). \quad (28)$$

There are at most $m + 1$ transitions per state. Constructing the graph and performing breadth-first search therefore take $O((m + 1)V^2)$ operations after constructing the profiles. Recording a shortest path recovers an explicit optimal inspection sequence. A finite optimum is at most $(E + 1)(M + 1) - 1$, since a shortest path repeats no state. The bounds are polynomial in the number of rooms, rather than in the logarithms of the side lengths.

Equivalently, all budgets can be considered together. Let $W_t(u, v)$ be the smallest daily budget sufficient in at most t days. Then

$$\begin{aligned} W_0(0, 0) &= 0, & W_0(u, v) &= \infty \quad ((u, v) \neq (0, 0)), \\ W_{t+1}(u, v) &= \min_{\substack{0 \leq i \leq u \\ 0 \leq j \leq v}} \max\{u + v - i - j, W_t(g_1(j), g_0(i))\}. \end{aligned} \quad (29)$$

This exact recurrence includes arbitrary interleaving of the two initial parity cohorts; it makes no serial-allocation assumption.

Remark 5.7. Every weight and lexicographic prefix is a checkerboard corner ideal: a distinct coordinatewise smaller vertex of the same parity has smaller weight. Hence the theorem also proves a corner-ideal normal form for the full-board game on every all-odd box. It does not assert that a global prefix can replace an arbitrary prescribed partial starting set without altering its optimum, and it does not settle boxes with even side lengths. The all-odd-box result gives exact feasibility, minimum time, and strategies through a polynomial recurrence; a uniform closed arithmetic expression for that time is a further question.

6 Paths: matching geometry and a probe-count potential

We prove the path formula stated in (1). The geometric lower proof applies to arbitrary possible-position sets. An interval description is used only for the explicit strategy attaining the bound.

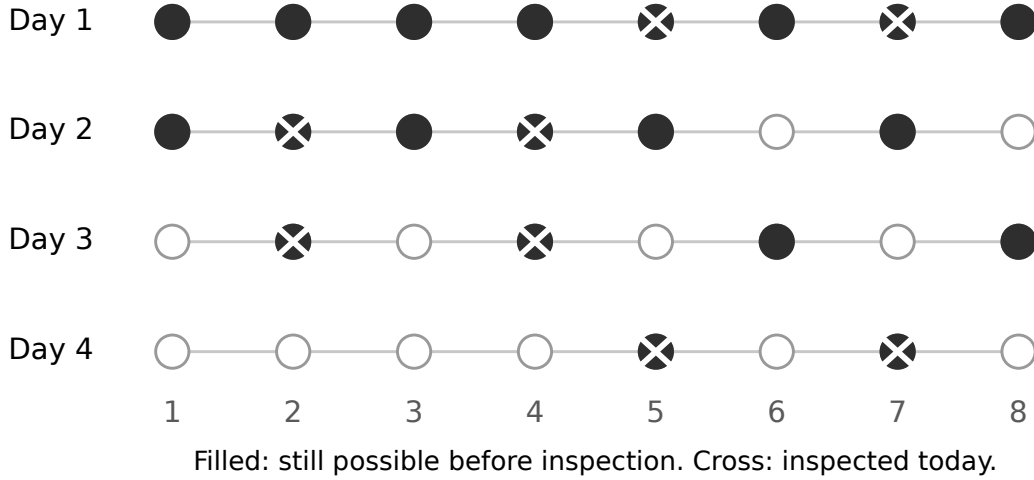


Figure 3: An optimal four-day search of eight rooms with two daily probes. The rows show every position still possible before that day's inspection.

6.1 Neighborhood inequalities

Number the vertices $0, \dots, n-1$. Each set below is contained in one color class. If n is even, the edges $(0,1), (2,3), \dots$ give a perfect matching and therefore $|N(R)| \geq |R|$. We need the following information about equality.

Lemma 6.1. *On an even path, if $0 < |R| < n/2$ and $|N(R)| = |R|$, then $N(R)$ contains neither endpoint. Every nonempty set containing neither endpoint has strictly more neighbors than vertices. On an odd path with larger class E and smaller class O ,*

$$|N(R)| \geq \min\{|R|, |O|\} \quad (R \subseteq E), \quad |N(R)| \geq |R| + 1 \quad (\emptyset \neq R \subseteq O).$$

Proof. Let M send each vertex of an even path to its matching partner. Equality implies $N(R) = M(R)$. If R consists of even vertices and contains $2i > 0$, its neighbor $2i-1$ forces $2i-2 \in R$. Thus

equality makes R an initial segment of the even vertices. Similarly an odd equality set is a final segment of the odd vertices. These statements prove both assertions about endpoints: a proper initial even segment has an odd neighborhood missing the far endpoint, and a proper final odd segment has an even neighborhood missing the other endpoint. Conversely, a nonempty equality set must itself contain an endpoint.

When n is odd, for every even vertex g there is a matching covering all vertices except g : match consecutive vertices separately to its left and right. If $R \subsetneq E$, choose $g \notin R$ to inject R into $N(R)$. For $R = E$ its neighborhood is O . Finally, for nonempty $R \subseteq O$, the matching exposing vertex 0 injects R into $N(R)$, and equality is impossible: the least occupied odd vertex has a left neighbor that would force a smaller occupied odd vertex, or the unmatched vertex 0. Thus there is at least one extra neighbor. $\square$

Track separately the targets that started in the two colors. Their current colors are opposite each day, so their allocations p, q always satisfy $p + q \leq m$. Write b for a cohort's size before inspection. On an even path retain a bit z : it is one just after a nonempty proper zero-growth neighborhood, and zero otherwise. Such a set is endpointless by the lemma. A surviving subset of it therefore expands strictly. The rank $u = 2b + z$, with rank zero for the empty set, is at most n and each physical step dominates

$$F_C(u, p) = \begin{cases} 0, & \lfloor u/2 \rfloor \leq p, \\ \min\{C, u - 2p + c\}, & \lfloor u/2 \rfloor > p, \end{cases} \quad (30)$$

with $C = n, c = 1$. Indeed, a zero-growth step raises the bit from zero to one; a step following bit one gains a vertex and may discard the bit. Both changes give at least one unit after subtracting $2p$.

For odd n , assign rank $2b$ in E , rank $2b + 1$ in nonempty O , and zero to the empty set. The same comparison holds with $c = 1$ and alternating new caps $n + 1$ in E and n in O . Initially the ranks are $n + 1, n$. Replacing both caps and initial ranks by n gives a valid lower comparison, since F_C is nondecreasing in C and u .

6.2 A general potential for the counting argument

Lemma 6.2 (Affine-rank potential). *Let $0 \leq c < 2m$, $D = 2m - c$, and $u \leq C$. Define*

$$J(u) = \left\lfloor \frac{\max\{u - c - 2, 0\}}{D} \right\rfloor, \quad \Phi(u) = \left\lfloor \frac{u + cJ(u)}{2} \right\rfloor.$$

For $0 \leq p \leq m$, the constant-cap transition (30) satisfies $\Phi(u) \leq p + \Phi(F_C(u, p))$. Consequently two such ranks sharing budget m need at least $\lceil (\Phi(u_0) + \Phi(v_0))/m \rceil$ rounds to become zero.

Proof. Both J and Φ are nondecreasing. If capture occurs, then $u \leq 2p + 1 \leq 2m + 1$, whence $J(u) = 0$ and $\Phi(u) \leq p$. If $2p \leq c$ and capture does not occur, the next rank is at least u , because $u \leq C$. Otherwise put $v = u - 2p + c$. It is positive and at most $u \leq C$, so the cap does not act. Since $2p - c \leq D$, $J(u) \leq J(v) + 1$. Therefore

$$u + cJ(u) \leq v + 2p + cJ(v).$$

Divide by two and take floors. Summing the resulting inequalities over both ranks and all rounds proves the last assertion. $\square$

For paths $c = 1$. Suppose $n > m \geq 2$, put $D = 2m - 1$, and use positive remainder coordinates

$$n = qD + r + 2, \quad 1 \leq r \leq D.$$

Here $n \geq 3$ and $q \geq 0$. Direct substitution gives

$$\Phi(n) = qm + \lfloor (r + 2)/2 \rfloor.$$

The resulting lower bound is $2q + 1$ when $r \leq m - 2$, or when $r = m - 1$ and m is even; otherwise it is $2q + 2$. For even n this is the required answer. For odd n it misses just the case m even and $r = m - 1$, which we now settle.

6.3 The equality obstruction on odd paths

Here $n = q(2m - 1) + m + 1$ with m, q even. Assume a strategy succeeds in $T = 2q + 1$ days. Use the exact alternating-cap comparison counters. For each counter, select its first zero and its last full state before that zero. If this interval starts at time a , has length L , and uses P probes, all intermediate transitions are unsaturated. Telescoping the affine transitions, including the final capture inequality, gives

$$C(a) + L \leq 2P + 2, \quad P \leq Lm.$$

Since $C(a) \geq n$, these imply $L \geq q + 1$ and $P \geq A := qm + m/2$. The two cohorts together have only $Tm = 2A$ probes, so equality holds for both. In particular $L = q + 1$ and $C(a) = n$: both intervals start when their cohort is in the smaller color class.

Their starting times have opposite parity and are at most q . The odd one, a , satisfies $1 \leq a \leq q - 1$ because q is even. If the even starting time is zero, both intervals lie in the first $2q$ days; otherwise both lie in the last $2q$ days. Their total probe count $2A = (2q + 1)m$ exceeds the $2qm$ available in either window. This contradiction proves that one extra day is necessary. The argument permits overlapping active intervals and arbitrary wasted inspections outside them.

For $m = 1$, the same potential has $D = 1$ and $\Phi(n) = n - 2$ for $n \geq 3$, giving $2n - 4$ days. On P_2 one probe cannot inspect both initial possibilities, and two consecutive inspections of one endpoint suffice. A single room is inspected directly.

6.4 An explicit strategy attaining the bound

For this construction number rooms $1, \dots, n$. Put

$$A(r) = \{r, r - 2, r - 4, \dots\} \cap \{1, \dots, n\}.$$

For $1 \leq r < n$, $N(A(r)) = A(r + 1)$. Inspecting the b largest vertices of $A(r)$ leaves $A(r - 2b)$; if nonempty, movement changes the frontier to $r - 2b + 1$. Reflection in the middle of the path gives the same rule in the opposite direction.

Assume $n \geq 3$ and $n > m$. Set $d = 2m - 1$ and

$$e = \left\lceil \frac{n - 2}{d} \right\rceil, \quad \rho = n - 2 - (e - 1)d, \quad c = \lfloor \rho/2 \rfloor + 1, \quad \ell = m - c.$$

Thus $1 \leq \rho \leq d$. Start with the cohort occupying $A(n - 1)$ and inspect its largest m possible rooms each day. Its frontier before day t is $n - 1 - (t - 1)d$, so it finishes on day e , using only c inspections

that day. If $\ell > 0$, assign the remaining ℓ inspections to the other cohort immediately; if $\ell = 0$, start that cohort on day $e + 1$.

Before its first inspection the second cohort is a whole color class. For odd n , use the same orientation. Its frontier is n on an odd starting day and $n - 1$ on an even starting day. For even n , choose the orientation whose frontier is $n - 1$: reflect when the starting day is odd. If its first batch has size b , the surviving frontier is $R_0 - 2b + 1$. Continue with batches of m until capture. A nonempty frontier $A(R)$ with $R \geq 2$ requires $\lceil (R - 1)/d \rceil$ such days; a singleton is inspected once.

For completeness the arithmetic determining the joining day is as follows. If $\ell = 0$ the total is $2e$. If $\ell > 0$, the totals are

$$2e - 2 + \left\lceil \frac{2\rho + 2}{d} \right\rceil \quad (n \text{ odd}), \quad 2e - 2 + \left\lceil \frac{\rho + 2c}{d} \right\rceil \quad (n \text{ even}).$$

They equal $2e - 1$ exactly when $\rho \leq m - 2$, or when $\rho = m - 1$ and n, m are both even; otherwise they equal $2e$. Comparing with $\lceil (2n - 4)/d \rceil$ proves (1). For $m = 1$ it gives $2n - 4$. If $n \leq m$, inspect every vertex on the first day. This completes both the construction and the arbitrary-strategy lower bound.

7 Two rows: an exact pair of counters

A two-row rectangle $P_2 \square P_n$ is also called a *ladder*: the two horizontal paths are joined by one vertical edge in each column. Its graph terminology does not impose any restriction on the searcher's movements between inspections.

For a cohort of fixed initial color, at any one time exactly one room per column has the right color. Identify its possible rooms with their column set $B \subseteq \{1, \dots, n\}$. A vertical move keeps the column, and a horizontal move changes it by one. Consequently the next column set after inspections is the *closed* path neighborhood

$$N_{P_n}[B \setminus S] = (B \setminus S) \cup N_{P_n}(B \setminus S).$$

Every nonempty proper column set gains at least one column. An initial or final interval attains the bound and stays an interval. Therefore the exact one-cohort size transition with p probes is

$$F(b, p) = \begin{cases} 0, & p \geq b, \\ \min\{n, b - p + 1\}, & p < b. \end{cases}$$

The two counters start at (n, n) and share the budget m . This is an exact lower and upper model: the inequality holds for arbitrary supports, and two boundary intervals attain every chosen sequence of allocations.

Theorem 7.1. *For $n \geq 2$ the minimum-time function is (2), with value ∞ for $m \leq 1$ and value one for $m \geq 2n$.*

Proof. For $2 \leq m < 2n$, put $N = n - 1$ and $M = m - 1$. Assign a counter of size b the weight $w(b) = \max\{b - 1, 0\}$. With a positive allocation p , its weight drops by at most $p - 1$; with no allocation its weight cannot drop. Thus the combined daily drop is at most M , and $T \geq \lceil 2N/M \rceil$.

Suppose $2N = TM$. Equality in this bound requires every day to allocate all m probes to exactly one cohort and to reduce its weight by exactly M . Each initial weight N must consequently be

divisible by M . If $2N$ is divisible by M but N is not, one extra day is necessary. This is exactly the exceptional condition that M is even and $N \equiv M/2 \pmod{M}$.

To attain the answer, write $N = qM + s$, $0 \leq s < M$. If $s = 0$, clear the first interval in q full days and the second in q more. Suppose $s > 0$. After q full days on the first interval its size is $s + 1$. On the next day finish it with $s + 1$ probes and assign the remaining $M - s$ to the other, still full, interval. Its next size is

$$b = n - (M - s) + 1 = (q - 1)M + 2s + 2.$$

It then needs $\lceil (b - 1)/M \rceil$ days. The total is

$$2q + \left\lceil \frac{2s + 1}{M} \right\rceil,$$

which is the claimed ceiling plus precisely the equality correction. The second interval always survives the shared day. If $q = 0$, then $s = n - 1$ and its allocation is $M - s = m - n < n$, because $m < 2n$. If $q \geq 1$, then $M \leq n - 1$ and $M - s < n$ as well. Thus $b \geq 2$ in both cases, and the displayed solo-day formula counts at least one remaining day. The algebra also holds for $q = 0$, by the integer translation rule for ceilings.

For one probe, an uninspected cohort remains full. Inspecting one room of a full color class also leaves the other color class full after movement, since every room on a ladder with $n \geq 2$ has at least two neighbors. Hence capture is impossible. The one-day assertion is immediate from the number of initial rooms. For $n = 1$ the graph is P_2 and the path result applies. $\square$

8 Three rows: a historical rank and the exact time

We prove the three-row classification in (3), including optimal inspection schedules. Put $G = P_3 \square P_n$. The length $n = 1$ is a three-vertex path. Henceforth $n \geq 2$; a four-cycle contained in G precludes capture with one inspection per day. Indeed, a target may stay on that cycle, where a nonempty parity cohort always has two possible vertices after movement. The constructions below show that two inspections suffice.

8.1 Even lengths: the efficient survivor sets

Let $n = 2s$ and $h = 3s$, so both parity classes have h vertices. In coordinates (column, row), label one parity by

$$a_i = (2i, 0), \quad b_i = (2i, 2), \quad c_i = (2i + 1, 1) \quad (0 \leq i < s).$$

Match these vertices respectively to $(2i + 1, 0)$, $(2i + 1, 2)$, $(2i, 1)$ in the other parity. Contracting an edge followed by the inverse matching map gives a directed graph D on the labelled parity. Besides loops, its arcs are

$$a_i \leftrightarrow c_i \leftrightarrow b_i, \quad a_i \rightarrow a_{i-1}, \quad b_i \rightarrow b_{i-1}, \quad c_i \rightarrow c_{i+1},$$

with out-of-range subscripts omitted. Consequently, for every survivor set R in this parity,

$$|N_G(R)| - |R| = |N_D^+(R) \setminus R|. \quad (31)$$

Each triple is strongly connected; its middle chain moves right and its outer chains move left. Thus D is strongly connected, and every nonempty proper parity set has surplus at least one.

Lemma 8.1. *If R is nonempty, $|R| \leq h - 2$, and $|N(R)| = |R| + 1$, then*

$$R = \{a_i, b_i, c_i : 0 \leq i < j\} \cup U, \quad U \subseteq \{a_j, b_j\}, \quad 0 \leq j < s.$$

For the opposite parity the analogous sets start at the other end. Moreover, if $R' \subseteq N(R)$ is nonempty, also has surplus one, and $|R'| \leq h - 2$, then $|R| = h - 2$ and $|R'| = 1$.

Proof. By (31), the outside boundary is a single vertex v , and R is closed under outgoing arcs of $D - v$. Deleting an outer vertex leaves a strongly connected graph: the other outer chain still moves left, the middle chain moves right, and all remaining outer vertices connect to their middle vertices. A nonempty closed set would then have all $h - 1$ remaining vertices, contrary to the hypothesis.

Deleting c_j leaves the prefix triples, the two singletons a_j, b_j , and the suffix triples as strongly connected components, omitting empty components. Arcs go from the suffix to each singleton and from each singleton to the prefix. A nonempty outgoing-closed set of size at most $h - 2$ is therefore precisely the stated prefix with some of the two singletons. Reflection in the long coordinate exchanges the physical parities and gives the opposite description.

Every nonempty classified set contains an outer corner at its starting end, and a set of size at least two contains both outer corners there. The neighborhood of R contains no outer corner at the opposite end unless $j = s - 1$. In that case the size bound permits at most one of a_j, b_j . Containing an opposite corner therefore forces $|R| = h - 2$ and supplies exactly one such corner. Hence R' must be a singleton. $\square$

8.2 A rank valid for arbitrary supports

We first assume $m < h - 2$. Track the two initial-parity cohorts separately. For a physical support A , set $z = 1$ if the preceding survivor set was nonempty, had surplus one, and had size at most $h - 2$; otherwise set $z = 0$. In particular full and empty supports have label zero. Define $\rho(A) = 2|A| + z$ for nonempty A , and $\rho(\emptyset) = 0$. This label records history and imposes no shape restriction on A .

For a numerical rank $0 \leq u \leq 2h$ and an allocation p , put

$$F_h(u, p) = \begin{cases} 0, & \lfloor u/2 \rfloor \leq p, \\ \min\{2h, u - 2p + 3\}, & \lfloor u/2 \rfloor > p. \end{cases} \quad (32)$$

It is nondecreasing in u and nonincreasing in p .

Lemma 8.2. *If $p \leq m < h - 2$ useful probes leave $R \subseteq A$, then $\rho(N(R)) \geq F_h(\rho(A), p)$.*

Proof. Write $r = |R| = |A| - p$. Empty survivors are immediate. If $r = h$ or $r = h - 1$, the neighborhood is full: every opposite-parity vertex has degree at least two. Thus its rank is the cap $2h$.

Suppose $1 \leq r \leq h - 2$. If the surplus is one, the next label is one. The old label cannot be one: Lemma 8.1 would require a support of size $h - 1$ to be reduced to one vertex, using $h - 2$ probes. Hence the new rank is $2(r + 1) + 1 = \rho(A) - 2p + 3$. If the surplus is at least two, the new rank is at least $2r + 4 \geq \rho(A) - 2p + 3$, since the old label is at most one. Capping can only weaken these inequalities. $\square$

Starting from $(2h, 2h)$, compare any actual schedule to the two-counter process

$$(u, v) \longmapsto (F_h(u, p), F_h(v, m - p)), \quad 0 \leq p \leq m.$$

Its terminal inspection condition is $\lfloor u/2 \rfloor + \lfloor v/2 \rfloor \leq m$. Use the actual two useful allocations; assign unused budget arbitrarily. Monotonicity and Lemma 8.2 keep the abstract ranks below the actual ranks throughout, so the numerical process captures no later than the actual schedule.

Conversely this process is physically realizable. Order each parity by increasing coordinate sum, breaking ties by increasing column. The two orders are

$$\begin{aligned} E &:: (0, 0), \quad ((2i - 2, 2), (2i - 1, 1), (2i, 0))_{i=1}^{s-1}, \quad (2s - 2, 2), (2s - 1, 1), \\ O &:: (0, 1), (1, 0), \quad ((2i - 1, 2), (2i, 1), (2i + 1, 0))_{i=1}^{s-1}, \quad (2s - 1, 2). \end{aligned}$$

Empty repeated blocks are omitted. Reading the neighbors of successive vertices in these lists gives, for every $k > 0$,

$$N(E_k) = O_{\min(h, k+1)}, \quad N(O_k) = E_{\min(h, k+2)}.$$

Here E_k, O_k denote prefixes of size k ; empty prefixes have empty neighborhood. Deleting a suffix of p vertices and moving therefore realizes (32), with phases $z = 0, 1$ respectively. A full cohort may choose either reflected orientation freely; reflection in the long coordinate exchanges physical parities. The two cohorts choose these orientations independently. Proper cohorts retain their orientation until they fill a parity class again or disappear. Thus the numerical process gives the exact time for $m < h - 2$.

8.3 Closed formula for the even case

Assume $2 \leq m < h$, put $W = 2h - 4$, $D = 2m - 3$, and write $W = qD + r$, $0 \leq r < D$. A solo cohort receiving m inspections every day clears in $\lceil W/D \rceil$ days: before capture its rank decreases by D , and a final rank of at most $2m + 1 = D + 4$ can be inspected. Sequential solo sweeps therefore give $2q$ days if $r = 0$, and $2q + 2$ otherwise.

For $r > 0$, an overlap saves one day precisely in the case appearing in (3). Give q full-budget days to the first cohort, whose rank becomes $r + 4$. On the shared day clear it with $p = \lfloor (r + 4)/2 \rfloor$ probes and use the other $m - p$ probes on the second full cohort. Its new rank minus four is

$$(q - 1)D + r + 2p.$$

It clears in q further days if $r + 2p \leq D$. Since W is even and D is odd, $r \equiv q \pmod{2}$; this inequality is exactly $m \geq r + 4 - (q \bmod 2)$. It also ensures $m - p \geq 2$, so the shared day starts a proper sweep without hidden saturation.

For the matching lower bound, take any successful numerical schedule. For each cohort, retain only its interval from the last full state before first capture to that capture. Probes before this interval can be discarded: an uninspected full cohort remains full, and the retained trace starts in exactly the same state. There is no later saturation within the retained interval. If it has ℓ inspection days and uses P probes, the $\ell - 1$ nonterminal moves each add three rank units. The final capture inequality yields

$$P \geq \left\lceil \frac{W + 3\ell}{2} \right\rceil, \quad \ell D \geq W. \quad (33)$$

The intervals may overlap arbitrarily; their probe counts still sum to at most the budget of the whole schedule. If $r = 0$, each has $\ell \geq q$ and $P \geq qm$, giving $T \geq 2q$. If $r > 0$, each has $\ell \geq q + 1$, whence

$$P \geq qm + \lceil (r + 3)/2 \rceil.$$

This rules out $2q$ days. A $(2q + 1)$ -day schedule requires $m \geq 2 \lceil (r + 3)/2 \rceil = r + 4 - (q \bmod 2)$, exactly the condition used by the shared-day construction. Extra interleaving cannot evade it.

8.4 The even high-budget exceptions

The lower simulation used $m < h - 2$, so its boundary must be checked separately. A perfect matching gives $|N(B)| \geq |B|$ for every room set, including mixed parities. If $m < h$, at least $2h - m > m$ rooms remain after the first movement; two-day capture is impossible. If $h \leq m < 2h$, inspect one fixed physical parity on each of two consecutive days. One day requires $m \geq 2h$.

For $m = h - 1$, $h \geq 6$, and for $m = h - 2$, $h \geq 9$, there is a three-day construction. Start a phase-zero sweep on one cohort with m probes; its next size is $h - m + 1$. Clear it on day two, giving the remaining probes to the independently oriented other full cohort. The latter then has $2h - 2m + 2 \leq m$ rooms, and day three clears it. The preceding two-day obstruction proves optimality.

The remaining case is $h = 6, m = 4$. After the first movement at least nine rooms remain: neither initial cohort can be eliminated, and any touched proper survivor cohort expands by at least one. With no useful probe the bound is immediate. The perfect matching then leaves at least $9 - 4 = 5$ rooms after the second movement, precluding capture on day three. Two consecutive two-day cohort sweeps, using four probes and then three, give four days. Finally $h = 3$ is the two-row board $P_2 \square P_3$: its two-probe time is four, by (2). These boundary values agree directly with (3).

8.5 Odd lengths and the extra parity constraint

Let $n \geq 3$ be odd and write $3n = 2h + 1$. By Theorem 4.1 and Corollary 4.4, the diagonal minimizing prefixes reduce the unrestricted game exactly to their two counts. Their neighborhood sizes are $\min(h, k + 1)$ on nonempty majority prefixes and $\min(h + 1, k + 2)$ on nonempty minority prefixes. Assign rank $2b$ to majority prefixes and $2b + 1$ to nonempty minority prefixes; empty supports have rank zero. Put $W = 3n - 4$. The full ranks are $W + 5$ and $W + 4$, and every nonterminal unsaturated transition is again $u \mapsto u - 2p + 3$. Saturation uses the full rank of the next physical parity, so the caps alternate.

Suppose $2 \leq m < h$ and write $W = qD + r$ with $D = 2m - 3$ and $0 \leq r < D$. Begin with the minority cohort, of rank $W + 4$. If $r = 0$, then q is odd; after its q -day solo sweep the untouched cohort is also in minority parity. Two such sweeps give $2q$ days. If $r > 0$, either full rank clears in at most $q + 1$ days, because $W + 1 \leq (q + 1)D$, giving $2q + 2$ days sequentially.

For a shared day, give the first, initially minority, cohort q full-budget days and then $p = \lfloor (r + 4)/2 \rfloor$ probes to finish it. At that day the other cohort's full rank is $W + 4 + e$, where $e = 1 - (q \bmod 2)$. Its next rank minus four is

$$(q - 1)D + r + e + 2p.$$

Since W is odd, $r \equiv e \pmod{2}$. Thus $r + e + 2p \leq D$ is exactly $m \geq r + 4$, giving the $(2q + 1)$ -day construction. This condition also ensures a proper, unsaturated departure of the second cohort on the shared day.

For the lower bound, extract each cohort's last-full active interval as above. If it starts in minority parity put $e = 0$, and otherwise put $e = 1$. Its length and probe count satisfy

$$P \geq \left\lceil \frac{W + e + 3\ell}{2} \right\rceil, \quad \ell D \geq W + e. \quad (34)$$

When $r = 0$, $\ell \geq q$ gives $P \geq qm$ for each cohort, proving $T \geq 2q$. When $r > 0$, $\ell \geq q + 1$ gives $P \geq qm + \lceil (r + 3)/2 \rceil$, ruling out $2q$ days. A total of $2q + 1$ days therefore requires $m \geq 2 \lceil (r + 3)/2 \rceil$. For even r this is the claimed $m \geq r + 4$. For odd r the only remaining possibility is $m = r + 3$.

That possibility is impossible by equality and parity. Here m and q are even. Equality in the total probe budget forces both active intervals to have length $q + 1$ and to start in minority parity; a longer interval or a majority start increases the right side of (34). Number days from zero. The two initial cohorts have opposite parity, so their starting times consist of an odd a and an even b , both at most q . Thus $1 \leq a \leq q - 1$. If $b = 0$, both intervals lie within $[0, 2q]$; if $b > 0$, both lie within $[1, 2q + 1]$. Either way their total probe count is at most $2qm$, contradicting the necessary $(2q + 1)m$. This establishes the extra parity constraint and proves the odd case of (3).

For completeness, a Hamilton path supplies a matching missing only one vertex, so $|N(B)| \geq |B| - 1$ for every room set. When $m < h$, the first movement leaves at least $2h - m > m$ rooms, excluding two-day capture. For $h \leq m < 3n$, inspect the entire minority physical parity on each of two consecutive days. One day requires all $3n$ rooms. Together with the four-cycle obstruction and the already treated lengths one and two, this completes every three-row parameter case.

9 Four rows: counting productive inspections

Let $G = P_4 \square P_n$ with $n \geq 4$, and put $h = 2n$. We prove (4). Smaller lengths are the path, two-row, and three-row families after exchanging coordinates. The key simplification is that all surplus-one survivor sets have exceptional cardinalities.

9.1 A directed matching contraction

Match rows zero and one, and rows two and three, in each column. Identify one parity with two layers of matched pairs. Contracting an edge followed by the inverse matching gives a directed graph with loops, bidirectional horizontal edges in both layers, and one directed vertical rung in each column. The rung directions alternate with the column. The other physical parity reverses the rungs. As before,

$$|N_G(R)| - |R| = |N_D^+(R) \setminus R|.$$

Both rung directions occur, so D is strongly connected.

Lemma 9.1. *Every nonempty proper parity set has positive surplus. If its surplus is one, its size is in $\{1, h - 2, h - 1\}$. Consequently every parity set R satisfies $|N(R)| \geq g_h(|R|)$, where*

$$g_h(k) = \begin{cases} 0, & k = 0, \\ 2, & k = 1, \\ h - 1, & k = h - 2, \\ \min\{h, k + 2\}, & \text{otherwise.} \end{cases} \quad (35)$$

Proof. If the outside boundary consists of a single vertex v , the survivor set is outgoing-closed in $D - v$. Deleting v splits one horizontal layer into two segments while the other layer stays intact. Every segment of length at least two contains both rung directions and is strongly connected to the intact layer. These vertices form a single core. A segment of length one can only be an endpoint singleton; there cannot be two such segments because $n \geq 4$. If no singleton remains, the graph after deletion is strongly connected, so a nonempty closed set has size $h - 1$. Otherwise the singleton is joined to the core by one directed rung. It is either a sink, allowing closed-set sizes $1, h - 1$, or a source, allowing sizes $h - 2, h - 1$.

This proves the surplus-one classification. Strong connectivity gives positive surplus for all other nonempty proper sets, hence surplus at least two there. A full parity has full neighborhood, establishing (35). The exceptional values are attainable: use a corner singleton, the complement of the two neighbors of an opposite corner, or a whole parity minus one vertex. Attainment of the bulk values follows from the diagonal prefixes used below. $\square$

Two inspections cannot succeed. A cohort of size at least $h - 1$ leaves at least $h - 3$ survivors, whose neighborhood still has at least $h - 1$ vertices by (35). Starting from h , this barrier persists for both cohorts.

9.2 The three-inspection endpoint

Give a cohort of size b the weight $w(b) = \max\{b - 3, 0\}$. With zero or one useful probe this weight cannot decrease, by (35). With two probes it decreases by at most one; the only positive possibility is a full cohort becoming size $h - 1$. With three probes it again decreases by at most one: survivor sizes from two through $h - 3$ gain at least two neighbors, and the near-full cases satisfy the same bound directly. If only one survivor remains, the old size is four and the old weight is one.

Every nonempty support after movement has at least two vertices, since the grid has minimum degree two. Thus three probes cannot eliminate both cohorts on the same day. A successful schedule has two distinct elimination days. On either day the eliminated cohort already had weight zero; the other receives at most one useful probe and cannot lose weight. Every other day reduces total weight by at most one. Since the initial total is $2(h - 3)$, a T -day schedule has $2(h - 3) \leq T - 2$, or

$$T_3(G) \geq 2h - 4. \quad (36)$$

To attain this, use diagonal prefixes, ordered by coordinate sum and then increasing column. Their neighborhoods are opposite-parity prefixes. Both phases have surplus two at every prefix size $2, \dots, h - 3$. A corner-starting singleton has two neighbors; the other phase's singleton has three. Reflection in the short coordinate exchanges physical parities, permitting either full cohort to choose the corner-starting phase independently.

With three probes daily, a solo prefix cohort decreases from h to four in $h - 4$ rounds. All those survivor sizes lie between two and $h - 3$. Since $h - 4$ is even, the four-vertex support is now in the corner phase. Leave its corner singleton on the next day and inspect its two neighbors on the last. This takes $h - 2$ days per cohort, attaining (36) in $2h - 4$ days.

9.3 A lower game with free holding

For larger budgets, pointwise minimum neighborhoods need not have compatible minimizing shapes. Instead make the lower comparison game strictly easier. Define

$$H_h(b, p) = \min\{b, g_h(\max\{b - p, 0\})\}. \quad (37)$$

The searcher may hold a cohort's count unchanged for free. The profile g_h is nondecreasing, so H_h is nondecreasing in b and nonincreasing in p . Comparing the same useful allocations with arbitrary physical supports, Lemma 9.1 inductively keeps the relaxed counts no larger. Every successful physical schedule therefore clears the relaxed game within the same time.

Replace any inspection which does not decrease a relaxed count by zero probes on that cohort; $H_h(b, 0) = b$ makes this valid. Trim a final allocation to the cohort's actual size. Consider its k

remaining *productive inspections*; these need not be consecutive, and may overlap the other cohort's productive inspections. Its count strictly decreases on each, and the last clears it.

Every nonterminal productive transition pays surplus two, with only two possible one-unit savings:

- (A) The first productive inspection uses two probes and sends h to $h - 1$ with surplus one.
- (B) The penultimate productive inspection leaves a singleton, whose neighborhood has size two. The final useful allocation is then exactly two.

There are no others. Survivor size $h - 1$ returns a full support and is unproductive. Survivor size $h - 2$ produces $h - 1$, productive only from the initial count h . A singleton produces two, and the next productive inspection clears it. The profile never produces count three. Let $A, B \in \{0, 1\}$ record the savings and P be the total useful probes on this cohort. Telescoping gives

$$P \geq h + 2(k - 1) - A - B. \quad (38)$$

If either saving occurs, some productive allocation is exactly two; if both occur, the first and last are distinct allocations of two. Hence respectively

$$P \leq (k - 1)m + 2, \quad P \leq (k - 2)m + 4. \quad (39)$$

Put $D = m - 2$, with $m \geq 3$. By monotonicity, spending m each day is best for a single relaxed cohort. The largest initial count it can clear in $j \geq 1$ inspections is

$$K_j = \min\{h, jD + 2\}. \quad (40)$$

For $j = 1$ this is direct capture. For the induction, if $b \geq m + 2$, its survivor size $b - m$ lies between two and $h - 3$ and the next count is $b - D$. If $b = m + 1$, it leaves a singleton and produces two rooms, already below the next threshold because $K_j \geq m \geq 3$. This proves (40). Thus each cohort needs at least $\lceil (h - 2)/D \rceil$ productive inspections, regardless of interleaving.

9.4 The residue lower bound

Assume $3 \leq m < h$, put $W = h - 2$, and write $W = qD + r$, $0 \leq r < D$. The case $m = 3$ is (36), so assume $m \geq 4$.

If $r = 0$, each cohort has $k \geq q$. If $k \geq q + 1$, (38) gives $P \geq qm$ even allowing both savings. If $k = q$, one saving would require $P \geq qm - 1$ but $P \leq qm - m + 2 < qm - 1$; two would require $P \geq qm - 2$ but $P \leq qm - 2m + 4 < qm - 2$. Therefore no savings occur and again $P \geq qm$. The two cohorts require $2qm$ probes, so $T \geq 2q$.

If $r = 1$, each has $k \geq q + 1$. For $k \geq q + 2$, (38) gives $P \geq qm + 3$. For $k = q + 1$, both savings would require $P \geq qm + 1$ and $P \leq qm + 4 - m \leq qm$, a contradiction. At most one occurs, so $P \geq qm + 2$. The combined $2qm + 4$ probes rule out $2q$ days, yielding $T \geq 2q + 1$.

Finally suppose $r \geq 2$. Again $k \geq q + 1$. If $k \geq q + 2$, (38) gives $P \geq qm + r + 2$. If $k = q + 1$, exactly one saving would require $P \geq qm + r + 1 > qm + 2$, contradicting the first inequality in (39). Two would require $P \geq qm + r \geq qm + 2$, contradicting the second, whose right side is $qm + 4 - m \leq qm$. Neither saving is possible, so again $P \geq qm + r + 2$. The combined probe total is at least $2qm + 2r + 4$. A $(2q + 1)$ -day schedule therefore requires $m \geq 2r + 4$; if this fails, $T \geq 2q + 2$. These are exactly the lower cases of (4), proved even with free holding.

9.5 Attaining schedules for every budget

For an actual diagonal prefix, when its survivor size is between two and $h - 3$, movement adds exactly two. A singleton has at most three neighbors, all inspectable next day because $m \geq 3$. Thus the solo capacity calculation (40) also proves an attaining sweep from a full cohort. Two sequential sweeps take $2q$ days for $r = 0$, and $2q + 2$ otherwise.

When $r = 1$, choose the first cohort's initial reflected orientation so that the singleton survivor on its q th full-budget day is a corner. Its next support has two rooms. On day $q + 1$, clear it with two probes and spend $m - 2 = D$ on the second full cohort. Since $W = h - 2$ is even, the remainder one cannot occur with $D = 2$; therefore $D \geq 3$. The second support now has size

$$h - D + 2 = (q - 1)D + 5 \leq qD + 2,$$

so q further full-budget days clear it. This attains $2q + 1$ days.

If $r \geq 2$ and $m \geq 2r + 4$, the first cohort has $r + 2$ rooms after q full-budget days. Clear it on the shared day with $r + 2$ probes and give $m - r - 2 = D - r \geq r + 2 \geq 4$ to the second cohort. This transition lies in the bulk, and its next size is

$$h - (D - r) + 2 = (q - 1)D + 2r + 4 \leq qD + 2.$$

It clears in q more days, again attaining $2q + 1$. Only the explicit compatible prefix families are used; there is no assumption that arbitrary minimum-neighborhood shapes can be chained.

The extreme budgets are immediate. A perfect matching gives $|N(B)| \geq |B|$ for arbitrary mixed-parity room sets. When $m < h$, the first movement leaves at least $2h - m > m$ rooms, ruling out two-day capture. For $h \leq m < 2h$, inspect one fixed physical parity on each of two consecutive days. One day requires all $2h = 4n$ rooms. The two-probe barrier proved above establishes infeasibility for all $m \leq 2$. This completes (4) and supplies an explicit optimal schedule in every feasible case.

10 Five rows

The first budget beyond the four-row classification admits another uniform formula. The two parities of the long side require different lower arguments.

Theorem 10.1 (Five rows, three inspections). *For every $n \geq 5$,*

$$T_3(P_5 \square P_n) = 10n - 20. \tag{41}$$

The odd-length proof uses the profiles of Theorem 4.1. The even-length proof uses an exhaustive finite symbolic certificate for arbitrary supports. Both arithmetic potential inequalities are verified in Lean; the complete five-row geometric argument is not presently formalized.

10.1 Odd lengths: exact profiles and a parity potential

Suppose $n \geq 5$ is odd, and put $h = (5n - 1)/2 \geq 12$. The even physical parity has $h + 1$ vertices and the odd physical parity has h . Denote these parities by $z = 0, 1$, respectively. Specializing

Theorem 4.1 to short-side parameter $b = 2$ gives $g_0(0) = g_1(0) = 0$ and

$$g_0(k) = \begin{cases} h, & h \leq k \leq h+1, \\ k+1, & k=1 \text{ or } h-3 \leq k \leq h-1, \\ k+2, & 2 \leq k \leq h-4, \end{cases} \quad (42)$$

$$g_1(k) = \begin{cases} h+1, & k=h, \\ k+2, & 1 \leq k \leq 2 \text{ or } h-2 \leq k \leq h-1, \\ k+3, & 3 \leq k \leq h-3. \end{cases} \quad (43)$$

For example, the complementary corner term in the majority profile is at most one exactly when $h+1-k \leq 4$. The minority corner term $Q(k)$ is at most two exactly when $k \leq 2$, and $Q(0) = 1$ handles its full set.

Define

$$W_h(b, z) = \begin{cases} 0, & b=0, \\ 1, & 1 \leq b \leq 2, \\ b-2, & 3 \leq b \leq 5, \\ \min\{2b+z-8, b+h+z-10\}, & b \geq 6. \end{cases} \quad (44)$$

A reachable nonempty support has at least three vertices in parity zero and at least two in parity one: these are the minimum degrees of vertices in the opposite parities. Thus its size satisfies

$$b=0 \quad \text{or} \quad 3-z \leq b \leq h+1-z. \quad (45)$$

Lemma 10.2 (Odd-length potential inequality). *For $h \geq 12$, $z \in \{0, 1\}$, a size satisfying (45), and $0 \leq p \leq 3$,*

$$W_h(b, z) \leq W_h(g_z((b-p)_+), 1-z) + \mathbf{1}_{\{p \geq 2\}}. \quad (46)$$

For fixed h, z , the function $W_h(b, z)$ is nondecreasing in b .

Proof. Substitute (42)–(44), split source and target sizes at 0, 2, 5, select the smaller affine expression for sizes above five, and split the survivor size at the displayed profile endpoints. Every resulting branch is a linear integer inequality under $h \geq 12$, $p \leq 3$, and (45). This complete case split, including truncated subtraction, is checked by `potential_step` and `value_monotone` in `FiveRowOddRankArithmetic.lean`. These proofs use Lean’s ordinary kernel-checked integer-arithmetic tactic and have no admitted cases. $\square$

Take p to be the number of useful probes actually meeting one cohort. Its survivor support has size $b-p$, and its actual next support has at least $g_z(b-p)$ vertices. Monotonicity makes (46) valid for this actual transition. The cohorts occupy disjoint parities each day. Their useful allocations sum to at most three, so at most one receives two or more inspections. The sum of their potentials decreases by at most one per day. Initially

$$W_h(h+1, 0) = W_h(h, 1) = 2h-9;$$

at capture both vanish. Thus every strategy takes at least $4h-18 = 10n-20$ days.

For an attaining strategy, sweep the initially minority cohort first. Use the compatible diagonal prefixes of Theorem 4.1; each day delete its last three vertices, or all of it if fewer remain. The first two transitions are

$$(h, 1) \longrightarrow (h, 0) \longrightarrow (h-2, 1).$$

For every minority size $6 \leq b \leq h - 2$, two rounds give

$$(b, 1) \longrightarrow (b, 0) \longrightarrow (b - 1, 1).$$

After $h - 7$ such pairs the support has size five in the minority parity. The final three transitions are

$$(5, 1) \longrightarrow (4, 0) \longrightarrow (2, 1) \longrightarrow 0.$$

The solo duration is $2 + 2(h - 7) + 3 = 2h - 9$, which is odd. The untouched cohort stays the full alternating parity, and is consequently minority when its own sweep begins. Repeat the same sweep, for a total of $4h - 18$ days. For $n = 5$, each solo sweep takes fifteen days.

10.2 Even lengths: a finite symbolic boundary classification

Suppose $n = 2q \geq 6$, and put $h = 5q$. Match horizontal pairs of columns. In row y and matched column j , the physical parity-zero vertex is $(2j + (y \bmod 2), y)$. Use the matching to identify the opposite parity with the same $5 \times q$ array. The resulting directed adjacency has a loop at every vertex, bidirectional vertical edges, leftward horizontal edges on rows $0, 2, 4$, and rightward edges on rows $1, 3$. The other physical parity reverses the horizontal arrows. For a support R , write $\sigma(R) = |N(R)| - |R|$. The matching identifies this surplus with its directed outside-boundary size. The directed graph is strongly connected: vertical travel reaches a row of either horizontal orientation. Thus a nonempty proper support has positive surplus.

Represent a column by a subset $B \subseteq \{0, 1, 2, 3, 4\}$, and let $V(B)$ be its vertical neighbors. For successive column masks A, B, C , define

$$c(A, B, C) = \left| \overline{B} \cap (V(B) \cup (A \cap \{1, 3\}) \cup (C \cap \{0, 2, 4\})) \right|. \quad (47)$$

The complement is within the five rows. With zero sentinel columns, the surplus of a word $B_1 \cdots B_q$ is exactly

$$\sum_{j=1}^q c(B_{j-1}, B_j, B_{j+1}).$$

For a bound $s \leq 2$, use states (A, B, e) , $0 \leq e \leq s$. Every $(0, B, 0)$ is initial. For each of the 32 choices of C , retain

$$(A, B, e) \longrightarrow (B, C, e + c(A, B, C))$$

if the new weight is at most s . A state is terminal for boundary exactly s when $e + c(A, B, 0) = s$. Explore every reachable state. Remove only self-loops, verifying that each repeats an empty or full column at zero additional weight. The remaining graph is acyclic. Its complete enumeration gives

	boundary one	boundary two
Reduced terminal paths	59	1057
Maximum reduced word length	5	8

At bound two the reduced reachable graph has 1124 states and 3874 edges. These counts follow from the stated finite rule; they are not a cutoff on physical grid length.

Every accepted word reduces to one of these terminal paths by deleting loop repetitions. Conversely, arbitrary nonnegative repetitions at the recorded positions preserve its boundary. A reduced

word with k present and a absent cells therefore represents counts $k + 5u$ and $a + 5v$, where $u, v \geq 0$ count allowed full and empty repetitions. A parameter is zero if its loop type is absent. Repetition preserves whether each row is a prefix.

Lemma 10.3 (Five-row small-boundary geometry). *For $q \geq 3$, in the directed orientation just specified:*

1. *A surplus-one support has size 1, $h - 2$, or $h - 1$.*
2. *A surplus-two support of size $3 \leq k \leq h - 5$ has left-prefix rows.*
3. *A surplus-two support with prefix rows and an empty row has at most five vertices.*
4. *A surplus-two prefix support of size four, five, or six has no neighbor in the last column.*

Finite symbolic verification. Enumerate every terminal path in the specified finite graph, recording every removed-loop position. For a reduced word let k, a be its present and absent counts, and let F, E indicate a full or empty repetition position. Discard only words shorter than three columns with no repetition positions. Every remaining word satisfies these checks:

1. At boundary one, either $k = 1$ and F is false, or $a \in \{1, 2\}$ and E is false.
2. At boundary two, F implies $k \geq 10$. If $(k \geq 3$ or $F)$ and $(a \geq 5$ or $E)$, all rows are prefixes.
3. At boundary two, prefix rows and an empty row imply that F is false and $k \leq 5$.
4. At boundary two, prefix rows and $4 \leq k \leq 6$ imply that F is false and

$$B_q \cup V(B_q) \cup (B_{q-1} \cap \{1, 3\}) = \emptyset.$$

These checks cover 59 and 1057 reduced words, respectively. They are implemented directly from (47) in `five_row_boundary_automaton.py`; the full words and repetition positions are included in the accompanying certificate. No interval assumption is made about arbitrary supports.

The first three conclusions persist for all repetition counts by $k \mapsto k + 5u$ and $a \mapsto a + 5v$. In the fourth size range, no full repetition is possible. Empty repetitions occur after the nonempty part of a prefix support, and cannot create a neighbor in the new last column. This proves the assertions for every $q \geq 3$. $\square$

Lemma 10.4 (Incompatible efficient transitions). *Suppose $\sigma(R) = 2$, $4 \leq k = |R| \leq h - 5$, and $R' \subseteq N(R)$ belongs to the opposite physical parity. If*

$$k - 1 \leq |R'| \leq k + 2, \quad |R'| \leq h - 5,$$

then $\sigma(R') \neq 2$.

Proof. If both surpluses were two, the preceding lemma would give left-prefix rows for R and right-prefix rows for R' , since the second parity reverses horizontal arrows. The neighborhood of a left-prefix support also has left-prefix rows. Its size is $k + 2 \leq h - 3$, so some row misses its last cell. The corresponding right-prefix row of R' must be empty. Part 3 gives $|R'| \leq 5$, hence $k \leq 6$. Part 4 now says $N(R)$ misses the entire last column, forcing every right-prefix row of R' to be empty, a contradiction. $\square$

10.3 The even-length historical potential and attaining sweeps

Mark a current cohort by $z = 1$ exactly when its immediately preceding survivor support had surplus two and size in $[4, h - 5]$; otherwise use $z = 0$. A marked support has $6 \leq b \leq h - 3$. Every nonempty reachable support has size at least two. Define

$$V_h(b, z) = \begin{cases} 0, & b = 0, \\ 1, & 1 \leq b \leq 2, \\ b - 2, & 3 \leq b \leq 5, \\ \min\{2b + z - 8, b + h - 10\}, & b \geq 6. \end{cases} \quad (48)$$

Here z records history, unlike the physical-parity variable in (44).

For $p \leq 3$ useful inspections put $r = b - p$, and denote the next state by (b', z') . These alternatives exhaust every physical transition:

1. $r = 0$, giving $(b', z') = (0, 0)$;
2. $r = h$, necessarily $b = h, p = 0$, giving $(h, 0)$;
3. $\sigma = 1$, giving $r \in \{1, h - 2, h - 1\}$ and $(b', z') = (r + 1, 0)$;
4. $\sigma = 2$, giving $b' = r + 2$ and $z' = \mathbf{1}_{\{4 \leq r \leq h-5\}}$; if $z = 1$ and $3 \leq r \leq h - 5$, this is forbidden by Lemma 10.4;
5. $\sigma \geq 3$, giving $b' \geq r + 3$ and $z' = 0$.

In the forbidden case, the previous survivor size is $k = b - 2$ and $r = k + 2 - p$, so $k - 1 \leq r \leq k + 2$ follows from $0 \leq p \leq 3$.

Substitution into (48) gives

$$V_h(b, z) \leq V_h(b', z') + \mathbf{1}_{\{p \geq 2\}}. \quad (49)$$

This is a finite split of linear integer inequalities under $h \geq 15$, $b \leq h$, $b = 0$ or $b \geq 2$, and $z = 1 \Rightarrow 6 \leq b \leq h - 3$. Its universal proof is `FiveRowRankArithmetic.potential_step` in the Lean artifact; the geometric alternatives remain the ordinary argument just given. The two-cohort potential decreases by at most one per day. Initially it is $2V_h(h, 0) = 4h - 20 = 10n - 20$, proving the lower bound.

For the upper bound use ascending $(x + y, x)$ prefixes, choosing a fixed reflected orientation independently for each initial cohort. Their surpluses are as follows; these describe the stated orders, rather than claiming that they simultaneously minimize all even-length profiles:

Phase	survivor sizes	surplus
0	$1, h - 2, h - 1$	1
0	all other proper nonempty sizes	2
1	$h - 1$	1
1	$1, 2, h - 4, h - 3, h - 2$	2
1	$3, \dots, h - 5$	3

Empty and full supports have empty and full neighborhoods. Neighborhoods of prefixes are opposite-phase prefixes. To derive the tables, let $L(d)$ be the length of diagonal $x + y = d$, and set $A(0) = 0$, $A(d + 1) = L(d) - A(d)$. For a prefix with j cells in its final diagonal, the surplus is

$$A(d) + 1 - \mathbf{1}_{\{d \geq 4\}} - \mathbf{1}_{\{\max(0, d-4) + j - 1 = n-1\}}.$$

The diagonal lengths are 1, 2, 3, 4, 5, then five through diagonal $n - 1$, then 4, 3, 2, 1. The recurrence gives $A(d) = \lceil d/2 \rceil$ for $d \leq 4$, $A(d) = 2 + (d \bmod 2)$ for $4 \leq d \leq n - 1$, and final values 2, 2, 1, 1. Substitution gives the tables and the neighborhood-prefix assertion.

Starting in phase zero, three rounds give $h \rightarrow h - 1 \rightarrow h - 2 \rightarrow h - 3$. Then $h - 8$ pairs of a stationary round and a one-cell decrease reach size five. The last three rounds are $5 \rightarrow 4 \rightarrow 2 \rightarrow 0$. The solo duration is

$$3 + 2(h - 8) + 3 = 2h - 10.$$

Reflection $x \mapsto n - 1 - x$ swaps physical parities because n is even, so each initial cohort can start its sweep in favorable phase zero. The untouched cohort remains full until its sweep begins. Two consecutive sweeps take $4h - 20$ days, completing Theorem 10.1.

10.4 Even lengths at every budget

The one-bit obstruction can be strengthened enough to resolve every budget on the even-length boards. For odd lengths, the exact two-count recurrence of Corollary 4.4 already supplies the answer at every budget. The following gives a closed expression on the remaining boards.

Theorem 10.5 (Five rows, even length, every budget). *Let $n \geq 6$ be even and $h = 5n/2$. Budgets one and two are impossible; $T_3 = 4h - 20$; and*

$$T_4 = 2 \left\lceil \frac{2h - 8}{3} \right\rceil. \quad (50)$$

For $5 \leq m < h$, write $2h - 6 = q(2m - 5) + r$, $0 \leq r < 2m - 5$, and set

$$J(0) = 0, \quad J(1) = 2, \quad J(2) = 3, \quad J(3) = J(4) = 4, \quad J(r) = \left\lceil \frac{r + 5}{2} \right\rceil \quad (r \geq 5).$$

Then

$$T_m = \begin{cases} 2q, & r = 0, \\ 2q + 1, & r > 0 \text{ and } 2J(r) \leq m, \\ 2q + 2, & \text{otherwise.} \end{cases} \quad (51)$$

Finally, $T_m = 2$ for $h \leq m < 2h$, and $T_m = 1$ for $m \geq 2h$.

We first strengthen the geometry in Lemma 10.3. Call the two endpoints in the last matched column *outward corners* when considering the reversed directed orientation.

Lemma 10.6 (Endpoint compatibility). *Suppose $\sigma(R) = 2$ and $3 \leq k = |R| \leq h - 5$. If $N(R)$ contains an outward corner, then $k \geq h - 6$. If $R' \subseteq N(R)$ has opposite physical parity, surplus two, and $3 \leq \ell = |R'| \leq h - 5$, then*

$$(k, \ell) \in \{(h - 6, 3), (h - 5, 3), (h - 5, 4)\}. \quad (52)$$

Proof. Write $n = 2s$. For $k \geq 6$, the prefix and empty-row conclusions of Lemma 10.3 imply that every row of R is a nonempty left prefix. Let its lengths be t_i and set $d_i = s - t_i$. The missing lengths in the rows of $N(R)$ are exactly

$$\begin{aligned} e_0 &= \min(d_0, d_1), & e_1 &= \min(d_0, (d_1 - 1)_+, d_2), & e_2 &= \min(d_1, d_2, d_3), \\ e_3 &= \min(d_2, (d_3 - 1)_+, d_4), & e_4 &= \min(d_3, d_4). \end{aligned}$$

The surplus identity is $\sum d_i = \sum e_i + 2$. Splitting the minima gives these three elementary integer consequences:

$$\begin{aligned} e_0 e_4 = 0 &\implies \sum d_i \leq 6, \\ (e_0 = e_1 = e_2 = 0) \text{ or } (e_2 = e_3 = e_4 = 0) &\implies \sum d_i \leq 5, \\ e_0 = e_2 = e_4 = 0 &\implies \sum d_i \leq 2. \end{aligned}$$

Their complete universal arithmetic proofs are the theorems `endpoint`, `three_rows`, and `alternating_rows` in `FiveRowDeficitArithmetic.lean`. The first proves the corner claim.

The rows of R' are right prefixes. Every occupied row must therefore be full in $N(R)$, that is, have $e_i = 0$. Since $|N(R)| \leq h - 3$, some row is not full, so the empty-row conclusion implies $\ell \leq 5$. The surplus-two prefix row vectors of sizes three through five, up to vertical reflection, are

$$\begin{array}{l|l} 3 & (2, 1, 0, 0, 0), (1, 1, 1, 0, 0), (1, 0, 1, 0, 1) \\ 4 & (2, 1, 1, 0, 0), (1, 1, 1, 0, 1) \\ 5 & (2, 1, 1, 0, 1), (1, 1, 1, 1, 1). \end{array}$$

This finite list follows by substituting the row-neighborhood maxima for vectors with total at most five. For $s \geq 6$ the far boundary cannot affect the check; direct substitution for $s = 3, 4, 5$ gives the same list. A size-three vector forces a zero endpoint deficit; a size-four vector forces three consecutive zeros at an end or the stronger alternating zeros; a size-five vector forces alternating zeros. The three inequalities above now give respectively $k \geq h - 6$, $k \geq h - 5$, and the contradiction $k \geq h - 2$.

For the source sizes $3 \leq k \leq 5$, use the same finite list directly. Every neighborhood row has length at most two, hence misses the last column because $s \geq 3$. It contains neither an outward corner nor a nonempty right prefix. This separate check is needed because the displayed deficit formula assumes that every source row is nonempty. $\square$

For this all-budget argument, mark a cohort by $z = 1$ when its previous survivor had surplus two and size in $[3, h - 5]$. Thus $z = 1$ implies $5 \leq b \leq h - 3$, and the previous survivor size was $b - 2$. Every physical transition, for p useful inspections and $a = b - p$, belongs to the following relation:

1. $a = 0$ gives $(0, 0)$, and $a = h$ gives $(h, 0)$;
2. surplus at least three gives any $(b', 0)$ with $a + 3 \leq b' \leq h$;
3. surplus one gives $(a + 1, 0)$ for $a \in \{1, h - 2, h - 1\}$, with $a = 1$ forbidden when $z = 1$ and $b < h - 4$;
4. surplus two gives $(a + 2, \mathbf{1}_{\{3 \leq a \leq h - 5\}})$; when $z = 1$ and $3 \leq a \leq h - 5$, the pair $(b - 2, a)$ must satisfy (52).

The last two restrictions are precisely Lemma 10.6. In particular, the large-surplus branch includes every possible target size, so this is a lower relaxation for arbitrary supports.

10.4.1 Uniform lower bound for $5 \leq m \leq h - 8$

Put $D = 2m - 5$, and for integer $u \geq 0$ define

$$\begin{aligned} j(u) &= \left\lfloor \frac{(u-7)_+}{D} \right\rfloor, & s(u) &= ((u-7)_+ \bmod D) + 1, \\ F_m(u) &= \left\lfloor \frac{u + 5j(u)}{2} \right\rfloor - \mathbf{1}_{\{j(u) \geq 1, s(u) \in \{1,2,4\}\}}. \end{aligned} \quad (53)$$

Its residue values give the following properties:

1. F_m is nondecreasing and $F_m(u+2) - F_m(u) \leq 3$;
2. if $\lfloor u/2 \rfloor \leq m$, then $F_m(u) = \lfloor u/2 \rfloor$;
3. if $p \leq m$ and $\lfloor u/2 \rfloor > p$, then $F_m(u) \leq p + F_m(u - 2p + 5)$.

For completeness, the last inequality without the indicator is the movement-five affine-ammunition inequality of Lemma 6.2. If $2p \leq 5$, monotonicity suffices. Otherwise the argument decreases by $2p - 5 \leq D$, so its quotient decreases by at most one. With no quotient decrease, the uncorrected inequality has slack at least two. With a quotient decrease, a discounted target has positive quotient and $s(u - 2p + 5) = s(u) + 2(m - p)$. Target remainder one, two, or four therefore forces source remainder one, two, or two/four, respectively. The source is also discounted, proving the inequality. The first two properties follow by substitution, including the first positive quotient and the wrap from remainder D to one.

Set

$$C = \min\{F_m(2h), 2 + F_m(2h - 2), 4 + F_m(2h - 4)\}, \quad \Phi(b, z) = \min\{C, F_m(2b + z)\}.$$

Every permitted transition lowers Φ by at most its useful allocation p . Here are all exceptional cases in that verification. A bulk surplus-two transition from $z = 0$ has target rank $u - 2p + 5$; from $z = 1$ it would require $p \geq h - 7 > m$ by (52). Surplus at least three has target rank at least $u - 2p + 5$. The three properties of F_m therefore handle these branches; a common cap preserves the inequality.

Capture follows from property 2. A surplus-one singleton is impossible from a marked state because it would require $p \geq h - 5$. From an unmarked state, $b \leq m + 1$; the only extra endpoint value is $F_m(2m + 2) = m + 2$, paid by m inspections and the target pair. For surplus two at survivor sizes one or two, states $b \leq m$ follow from cardinality. The four extra source values at $(b, z) = (m + 1, 0), (m + 1, 1), (m + 2, 0), (m + 2, 1)$ are $m + 2, m + 3, m + 4, m + 4$, respectively, and the target size is at most four.

Finally consider survivor sizes $h - 4, \dots, h - 1$. Rank increases need only monotonicity. The sole proper-source rank decrease is unmarked $h - 1 \rightarrow h - 2$ with three inspections, paid by property 1. The improvements from the full state to $h - 1$ with two inspections and to $h - 2$ with four inspections are exactly the last two terms defining C . The full, zero-inspection transition is tautological. This exhausts the relation.

Both initial cohorts have potential C and both final potentials vanish, so $T_m \geq \lceil 2C/m \rceil$. Direct substitution in the quotient and remainder of (51) gives, for $m \geq 6$,

$$C = qm + J(r) - \mathbf{1}_{\{r=2\}}.$$

When $m = 5$, the physical value h is divisible by five, so $r = 4$ and the same identity holds. The otherwise exceptional residues one and three at this budget cannot occur. At $r = 2$, $m \geq 6$ ensures that $\lceil 2C/m \rceil = 2q + 1$; all other residues immediately give the right side of (51).

10.4.2 The four-inspection lower bound

Define

$$H(b, z) = \begin{cases} b, & b \leq 4, \\ 6, & (b, z) = (5, 0), \\ 4 \lfloor (2b + z)/3 \rfloor - 8 + 3\mathbf{1}_{\{(2b+z) \bmod 3=2\}}, & \text{otherwise,} \end{cases}$$

and

$$A_1 = \min\{H(h-1, 0), 3 + H(h-2, 0)\}, \quad C_4 = \min\{H(h, 0), 2 + A_1, 4 + H(h-2, 0)\}.$$

Use potential C_4 at the full state, A_1 at the unmarked state $h-1$, and H elsewhere. Allocating $p \leq 4$ lowers it by at most p . This assertion is a finite residue split with the unbounded parameter $h \geq 15$: the theorem `potential_step` in `FiveRowFourProbeArithmetic.lean` checks every branch of the physical relaxation above. In this budget range a marked source forbids all surplus-one transitions and all bulk surplus-two transitions; the endpoint restrictions prove these exclusions. The checker includes all larger targets in the surplus-at-least-three branch.

Writing $h = 3t, 3t + 1, 3t + 2$ gives respectively

$$(A_1, C_4) = (8t - 12, 8t - 10), \quad (8t - 9, 8t - 8), \quad (8t - 5, 8t - 4).$$

For $L = \lceil (2h - 8)/3 \rceil$, this means $C_4 = 4L$ except when $h \equiv 0 \pmod{3}$, where $C_4 = 4L - 2$. The total-ammunition inequality already proves $T_4 \geq 2L$ outside that residue.

In the exceptional residue, $2L - 1$ days would use exactly $2C_4$ probes, so every cohort transition must be tight and every day must use four useful probes. From a full state, the only positive tight allocation is two inspections, producing size $h - 1$. Thus the first day must split $2 + 2$. From an unmarked state of size $h - 1$, every tight allocation requires four inspections. Two such transitions cannot share day two. This proves $T_4 \geq 2L$. The two local tightness assertions are universally checked by `full_tight` and `penultimate_tight` in `FiveRowFourProbeEquality.lean`.

10.4.3 High budgets and explicit attaining schedules

For $h \geq 25$ and $h - 7 \leq m < h$, write $s = h - m \in \{1, \dots, 7\}$. Then $q = 1$, $r = 2s - 1$, and the seven thresholds $2J(r)$ are 4, 8, 10, 12, 14, 16, 18, all at most $h - s$. Thus the proposed value is three. The matching forbids two days for every $m < h$: after day one at least $2h - m > m$ positions remain.

Only $h = 15, 20$ require additional lower checks. The universal cardinality profile is

$$g(0) = 0, \quad g(1) = 2, \quad g(k) = k + 2 \quad (2 \leq k \leq h - 3), \quad g(h - 2) = h - 1, \quad g(h - 1) = g(h) = h.$$

Iterate $(a, b) \mapsto (g((a - p)_+), g((b - m + p)_+))$, allowing every $0 \leq p \leq m$, sorting the pairs and retaining Pareto-minimal members. Monotonicity of g justifies every deletion. From (h, h) the required final antichains are

(h, m)	days played	remaining pairs
(15, 8)	3	(0, 11), (2, 10), (4, 9), (7, 8)
(15, 11)	2	(0, 12), (2, 11), (4, 10), (5, 9), (6, 8), (7, 7)
(20, 14)	2	(0, 16), (2, 15), (4, 14), (5, 13), (6, 12), (7, 11), (8, 10), (9, 9).

Every remaining total exceeds its budget, proving lower times five, four, and four. These finite universal allocation checks are kernel checked in `FiveRowHighBudgetArithmetic.lean`. Budget monotonicity covers $h = 15, m = 9, 10$ and $h = 20, m = 13$. All other high-budget lower bounds are three.

For the matching upper bounds, use the previously displayed physical prefix tables. With four inspections, start each solo sweep in phase one. Its first round gives $(h - 2, 0)$. At phase-zero size $b \geq 9$, two rounds give $(b - 2, 1)$ and $(b - 3, 0)$. Repeat until size 7, 8, 6, for h modulo three equal to 0, 1, 2, respectively. The final tails are $7 \rightarrow 5 \rightarrow 3 \rightarrow 0$, $8 \rightarrow 6 \rightarrow 4 \rightarrow 0$, and $6 \rightarrow 4 \rightarrow 0$. The solo duration is L , so two independently oriented sweeps attain $2L$.

For $m \geq 5$, full allocations decrease the bulk rank by $D = 2m - 5$. If $r = 0$, each solo sweep lasts q rounds. If $r > 0$, after q full allocations its remaining size is $J(r)$; substitution in the last two prefix transitions gives the special residues one, two, and four. A fresh cohort can be placed within q further rounds of capture by an initial allocation of $J(r)$. Indeed, the optimal oriented initial rank decreases at allocations $p = 0, 1, 2, 3, 4$ are 0, 0, 1, 2, 4, and the decrease is $2p - 5$ for $5 \leq p < h/2$. At $p = J(r)$ it is at least r , leaving rank at most $qD + 6$, which clears within q further full allocations by the same prefix table. The shared-round construction only needs $p = J(r) \leq m/2 < h/2$, within this range. Share the finishing and starting rounds exactly when $2J(r) \leq m$; otherwise take two separate solo sweeps. This attains (51).

The impossibility at budgets below three follows from the 4×4 subgrid obstruction. At budget at least h , inspect one full parity then the other cohort's full parity; one day is possible exactly at budget $2h$. Together with Theorem 10.1, these facts complete the proof of Theorem 10.5. The geometry here is an ordinary proof with a finite symbolic certificate; the cited Lean modules verify its universal scalar consequences, not the complete physical five-row classification.

11 Minimum-budget searches on seven odd widths

The exact profile recurrence can sometimes be reduced further to an affine formula in the rectangle's length. We prove such formulas at the minimum feasible budget for seven widths. The proof uses a finite rational certificate at each width together with an insertion lemma that covers every larger length.

Theorem 11.1. *For $w \in \{3, 5, 7, 9, 11, 13, 15\}$ and every odd $n \geq w$,*

$$T_{(w+1)/2}(P_w \square P_n) = 2wn - C_w, \quad \begin{array}{c|cccccc} w & 3 & 5 & 7 & 9 & 11 & 13 & 15 \\ \hline C_w & 8 & 20 & 44 & 84 & 136 & 208 & 288. \end{array}$$

The budget $(w + 1)/2$ is the minimum feasible budget.

The width-three and width-five entries agree with the separate formulas proved for those families. This theorem does not assert a formula for every odd width or optimality of serial allocation at other budgets. The certificates in this section are checked with exact rational arithmetic; they are not Lean proofs.

11.1 A separable lower-bound certificate

Set $w = 2b + 1$, $m = b + 1$, $D = m + 1$, and let the two color classes have sizes H and $H - 1$. Use the profiles g_0, g_1 from Theorem 4.1. Choose nonnegative rational numbers $c_p(r)$, for $p \in \{0, 1\}$

and $0 \leq r \leq m$, with

$$c_0(r) + c_1(m - r) \leq 1. \quad (54)$$

Suppose nonnegative functions F_p , with $F_p(0) = 0$, satisfy

$$F_p(k) - F_{1-p}(g_p((k - r)_+)) \leq c_p(r) \quad (0 \leq k \leq H - p, \ 0 \leq r \leq m). \quad (55)$$

Here $(x)_+ = \max(x, 0)$, as elsewhere in the search recurrences.

At a current state (u, v) the sum $F_0(u) + F_1(v)$ decreases by at most one per day, by (54). It suffices to consider strategies using the full quota whenever more than m possibilities remain: additional suffix inspections preserve the prefix normal form and cannot hurt. On the final day one can pad the two quotas to sum to m , with the positive-part convention making both survivors empty. Consequently

$$T_m \geq \lceil F_0(H) + F_1(H - 1) \rceil. \quad (56)$$

The functions are indexed by current color, so this argument includes arbitrary interleaving of the two initial cohorts.

For each finite certificate below, $F_p(k)$ is computed as the shortest total charge of a path from (p, k) to an empty cohort, where a quota r costs $c_p(r)$ and sends the cohort to $(1 - p, g_p((k - r)_+))$. Multiplying by the common denominator gives a finite graph with nonnegative integer edge weights. After computing the functions, the verifier checks every inequality (54)–(55) directly with exact fractions.

11.2 Inserting an arbitrarily long affine interval

Lemma 11.2 (Affine insertion). *Let $A = b(b + 1)$. Suppose a certificate at majority size H has an integer cut c such that*

$$c \geq A + 2D, \quad H - c \geq A + 2D + 2, \quad (57)$$

$$F_p(k) = 2k + \alpha_p \quad (c - 2D \leq k \leq c + 2D, \ p \in \{0, 1\}). \quad (58)$$

For every integer $\Delta \geq 0$, replace H by $H + \Delta$ in the profile formulas and define

$$F'_p(k) = \begin{cases} F_p(k), & k \leq c, \\ F_p(c) + 2(k - c), & c \leq k \leq c + \Delta, \\ F_p(k - \Delta) + 2\Delta, & k \geq c + \Delta. \end{cases} \quad (59)$$

Then F'_p satisfies the same charge inequalities. The initial lower bound (56) increases by exactly 4Δ .

Proof. The pieces of (59) agree at their endpoints. For either the base or enlarged profiles, every successor $j = g_p((k - r)_+)$ satisfies

$$|j - k| \leq D. \quad (60)$$

For a nonempty survivor, its surplus lies between -1 and $b + 1$, and $r \leq m$. For an empty survivor, $k \leq m$ and $j = 0$.

If $k < c - D$, the enlarged profile agrees with the base profile at $q = (k - r)_+$. Indeed, the far-corner terms in both profile formulas exceed their plateau caps by (57). The successor is unchanged and is less than c by (60). Both potentials are unchanged, so the base inequality applies.

If $k > c + \Delta + D$, write $k_0 = k - \Delta$ and $q_0 = k_0 - r$. Since $q_0 > c + D - m = c + 1$, the low-corner terms have reached their plateau caps. The profile formulas therefore give

$$g'_p(q_0 + \Delta) = g_p(q_0) + \Delta.$$

Writing j_0 for the base successor of k_0 , we have $j = j_0 + \Delta$ and $j_0 > c$. Both potentials gain 2Δ , which cancels in their difference.

Finally suppose $c - D \leq k \leq c + \Delta + D$. The survivor $q = k - r$ is positive, with

$$q \geq c - D - m \geq A + 1, \quad H + \Delta - q \geq H - c - D \geq A + D + 2.$$

Both profiles are therefore on their affine plateaus: $j = k - r + \delta_p$, where $\delta_0 = b$ and $\delta_1 = b + 1$. By (60), both k and j lie in $[c - 2D, c + \Delta + 2D]$. Throughout that interval the corresponding potentials are $2k + \alpha_p$ and $2j + \alpha_{1-p}$. Their difference is

$$2r - 2\delta_p + \alpha_p - \alpha_{1-p}.$$

This is exactly the base potential difference at source $k = c$ with the same current color and quota r . Its survivor and successor lie in the same affine profile and potential collar, so the verified base inequality bounds the expression by $c_p(r)$.

These three cases exhaust the enlarged state space. At each full color class (59) adds 2Δ to the potential, giving the stated increase of 4Δ in the integer lower bound. $\square$

The matching upper construction has the same insertion property. A *serial strategy* directs the full budget to one initial cohort until it can be eliminated, uses any unused quota on that finishing day on the other cohort, and then directs the full budget to the other.

Lemma 11.3 (Insertion in a serial strategy). *Under (57), suppose a base serial strategy visits count c in each active cohort before an inspection. An enlarged rectangle with majority size $H + \Delta$ has a strategy taking exactly 4Δ more days.*

Proof. Under a full-budget inspection an active cohort's size never increases, because every neighborhood surplus is at most m . The certified visit to c is therefore reached from above, and the subsequent base trajectory stays at or below c . Above the cut, the enlarged active trajectory is the translation by Δ of the base trajectory, by the high profile translation identity in the preceding proof. The uninspected cohort remains its full current color class. When the first active cohort reaches $c + \Delta$, insert 2Δ full-budget days. Throughout this inserted interval the profile plateaus give

$$g_0(k - m) = k - 1, \quad g_1(k - m) = k.$$

Every two days reduce its size by one and restore its current color. The inserted block therefore ends at c with the phase of the base schedule unchanged. Follow the base trajectory below the cut.

The first cohort's finishing-day unused quota is unchanged. The second cohort consequently begins its active phase at the translated base state. This state is above the cut: it lies within $m + 1$ of a full color class, whereas (57) places c farther from that edge. Follow its translated trajectory until $c + \Delta$, insert another 2Δ full-budget days, and then finish along the unchanged lower trajectory. Both inserted blocks have even length, so the later phases of the schedule agree with those in the base strategy. $\square$

The two insertion proofs also apply to any other compatible profile family with color-class sizes $H, H - 1$ and a fixed budget m , provided that a corner radius A has the following properties: every surplus is between -1 and m ; the central profiles are $k + m - 1$ and $k + m$; the low profiles are unchanged when the remaining tail exceeds A ; and the high profiles translate when the source count exceeds A . These are exactly the properties used in the displacement, collar, and trajectory arguments. In that formulation one sets $D = m + 1$ and uses the same margin and affine-collar conditions. This observation will also give an exact three-dimensional family below.

11.3 The finite certificates

The certificate parameters are as follows; in each row $\alpha_1 = \alpha_0 + 1$.

w	m	n_0	H	c	α_0
3	2	13	20	10	-4
5	3	13	33	16	-8
7	4	15	53	26	-15
9	5	15	68	34	-26
11	6	17	94	47	-241/6
13	7	19	124	62	-59
15	8	21	158	79	-80

The charge sequences, listed from quota zero to quota m , are

w	c_0	c_1
3	(0, 0, 1)	(0, 0, 1)
5	(0, 0, 1, 1)	(0, 0, 1, 1)
7	(0, 0, 0, 1, 1)	(0, 0, 1, 1, 1)
9	(0, 0, $\frac{1}{3}, \frac{2}{3}, 1, 1$)	(0, 0, $\frac{1}{3}, \frac{2}{3}, 1, 1$)
11	(0, 0, $\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, 1, 1$)	(0, 0, $\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, 1, 1$)
13	(0, 0, $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1, 1$)	(0, 0, $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1$)
15	(0, 0, $\frac{1}{7}, \frac{2}{7}, \frac{1}{2}, \frac{5}{7}, \frac{6}{7}, 1, 1$)	(0, 0, $\frac{1}{7}, \frac{2}{7}, \frac{1}{2}, \frac{5}{7}, \frac{6}{7}, 1, 1$)

Proof of Theorem 11.1. The exact verifier constructs the rational potentials for every odd n between w and the displayed n_0 , inclusive: 32 finite rectangles in total. For each it checks all charge and potential inequalities, and the equality

$$\lceil F_0(H) + F_1(H - 1) \rceil = \text{the displayed serial strategy's length} = 2wn - C_w.$$

At width eleven the unrounded bound is $2wn - 410/3$; its ceiling gives $2wn - 136$. The other six widths give the integer bound directly. At each base length n_0 , the verifier also checks every value in the affine collar (58), the margins (57), and both serial visits to the cut. The first initial cohort is the minority cohort except at widths seven and thirteen, where it is the majority cohort.

For any larger odd n , put $\Delta = w(n - n_0)/2$. The two insertion lemmas increase both the lower bound and the explicit strategy length by $4\Delta = 2w(n - n_0)$. This proves the formula for every larger length, while the finite certificates cover the remaining lengths. Finally, the majority profile's maximum surplus is b , so Corollary 5.6 gives the minimum feasible budget $b + 1 = (w + 1)/2$. $\square$

The exact checking program is `src/odd_minimum_budget_all_lengths.py` in the companion artifact. Its receipt `research/odd-minimum-budget-all-lengths.json` contains all base potential arrays and the checked quantities. The final verifier uses only integer and rational arithmetic. A numerical linear program helped discover the charge sequences, but is neither called nor trusted by this verifier. The finite checks establish the explicit premises of the two insertion lemmas; those lemmas, rather than an extrapolation of a numerical pattern, supply the infinite-length claim.

12 Eventual affine periods in odd cylinders

The interior-crossing argument extends to every fixed all-odd cross-section, in every dimension. The first step is an exact finite description of its two end regions; an affine plateau alone would not justify changing the cylinder's length.

Let Q be a fixed nontrivial box with odd side lengths. Put

$$W = |Q|, \quad S = \sum_i (\text{side}_i - 1), \quad Z = W(S + 1), \quad \beta = (W - 1)/2.$$

Consider $Q \square P_n$ with odd longitudinal length n . The one-point cross-section gives paths, already classified separately. For $n \geq S + 3$, the longitudinal coordinate is longest and is placed last in the order of Theorem 5.1. Write $H = (Wn + 1)/2$ for its majority class size and g_p^n for its exact color- p profile.

12.1 Two finite tables describe both ends

For $v \in Q_p$, let $r_p(v)$ be the rank of $(v, 0)$ in the parity order of the infinite prism $Q \times \{0, 1, \dots\}$. Every such point has weight at most S , so every preceding point has longitudinal coordinate at most S . Its rank is therefore independent of n for $n \geq S + 1$ and can be computed in the single finite slab $Q \square P_{S+1}$. At most $Z = W(S + 1)$ vertices have weight at most S , so $r_p(v) \leq Z$.

Using the transverse cost a_Q from (21), define

$$U_p(k) = \sum_{\substack{v \in Q_p \\ r_p(v) \leq k}} a_Q(v), \quad B_p(k) = |\{v \in Q_p : r_p(v) \leq k\}|.$$

Both tables are constant for $k \geq Z$.

Lemma 12.1 (Stable corner decomposition). *For $n \geq S + 3$ and $k > 0$,*

$$g_p^n(k) = k + \mathbf{1}_{\{p=1\}} + U_p(k) - |Q_p| + B_p(H - p - k), \quad g_p^n(0) = 0. \quad (61)$$

In particular, every surplus is at most $\beta + p$, and

$$k > Z, \quad H - p - k > Z \implies g_p^n(k) = k + \beta + p. \quad (62)$$

Proof. The ambient cost $a(x) - 1$ equals $a_Q(v)$ on the bottom slice, zero on an interior slice, and -1 on the top slice. This includes the origin: its ambient cost minus one is the transverse origin cost. The prefix-cost identity therefore counts the selected bottom weights through $U_p(k)$ and subtracts the number of selected top vertices.

Coordinate complement preserves parity and reverses both weight and lexicographic order. It carries the top slice to the bottom slice. Thus selected top vertices correspond to bottom vertices

outside the prefix of size $H - p - k$, and their number is $|Q_p| - B_p(H - p - k)$. This proves (61). The empty prefix is separate because the extra origin-neighbor term requires a nonempty odd prefix.

The transverse full-class cost identity gives $U_p(\infty) = |Q_{1-p}| - \mathbf{1}_{\{p=1\}}$ and $B_p(\infty) = |Q_p|$. Nonnegativity of a_Q gives the asserted upper bound on surplus. Stabilizing both tables gives (62). $\square$

First slice	Every interior slice	Last slice																											
<table><tr><td>2</td><td>1</td><td>0</td></tr><tr><td>2</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	2	1	0	2	1	0	1	1	0	<table><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	0	0	0	0	0	0	<table><tr><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>-1</td><td>-1</td><td>-1</td></tr></table>	-1	-1	-1	-1	-1	-1	-1	-1	-1
2	1	0																											
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Local contribution to neighborhood surplus in a $3 \times 3 \times n$ odd cylinder.

Figure 4: The local contribution $a(x) - 1$ in a cylinder with 3×3 cross-section: transverse costs on the first slice, zero in interior slices, and -1 on the last slice. The surplus of a nonempty odd prefix also has the additional 1 in (61); that term is outside the local sum.

The decomposition gives the stronger identities needed to change length. For two valid majority sizes H and $H + \delta$, with both longitudinal lengths at least $S + 3$, we have

$$H - p - k > Z \implies g_p^{H+\delta}(k) = g_p^H(k), \quad (63)$$

$$k > Z \implies g_p^{H+\delta}(k + \delta) = g_p^H(k) + \delta. \quad (64)$$

In the first identity the tail table is full; in the second the bottom weight table is full and the tail argument is unchanged. The first also holds at $k = 0$, since both profiles are zero there. Superscripts now indicate majority size rather than longitudinal length.

An all-odd box has a spanning path whose two endpoints lie in its majority color: traverse successive slices alternately forward and backward and induct on dimension. On that odd path a proper majority k -set has at least k neighbors, by omitting an unselected majority vertex and matching selected vertices toward it from both sides. A nonempty minority k -set has at least $k + 1$ neighbors, by its consecutive blocks in the spacing-two path order. The box contains these path edges, so

$$g_0(k) \geq k \ (k < H), \quad g_1(k) \geq k + 1 \ (k > 0), \quad g_0(H) = H - 1. \quad (65)$$

Corollary 12.2 (Eventual feasibility threshold). *If $H \geq 2Z + 3$, then $h(Q \square P_n) = (W + 1)/2$.*

Proof. The size condition implies $n > S + 3$. The majority surplus is at most β and attains β at $k = Z + 1$ by (62). Apply Corollary 5.6. $\square$

This is an eventual threshold; shorter cylinders can need fewer inspections. For example, a $5 \times 5 \times 5$ box needs twelve, whereas a sufficiently long cylinder with 5×5 cross-section needs thirteen.

12.2 The period and an explicit threshold

Fix a feasible eventual budget $m \geq \beta + 1$ and set

$$d = 2m - W > 0, \quad g = \gcd(W, d), \quad \Delta = \text{lcm}(W, d) = Wd/g.$$

For the explicit threshold define

$$\begin{aligned} J &= m + 1, & A &= Z + 1, & R &= \Delta + 2J, \\ B &= 8Z + 6d + 2, & L &= Z + m + 1, & \mathcal{E} &= B + 4L(B + 1), \\ H_* &= R + 2A + 4J + 2 + 2\mathcal{E}(R + 2J + 1). \end{aligned} \tag{66}$$

Theorem 12.3 (Eventual affine period in every odd cylinder). *For every odd n with $(Wn + 1)/2 \geq H_*$,*

$$T_m(Q \square P_{n+2d/g}) = T_m(Q \square P_n) + 4W/g. \tag{67}$$

At the eventual minimum budget $m = (W + 1)/2$, increasing a sufficiently large odd length by two increases the optimal time by exactly $4W$.

The threshold is not intended to be sharp. The proof uses the profile bounds, plateau and stable translations just proved, together with connectedness.

12.3 A potential with a bounded total deficit

For $n \geq S + 3$, write $H = (Wn + 1)/2$ and $M = H - 1$. Define

$$K = 2Z + 2m + 1, \quad \Gamma = (2H - 4Z - 2m - 5)_+, \quad F_p(k) = \min\{\Gamma, (2k + p - K)_+\}.$$

Lemma 12.4 (Uniform potential and time bounds). *For $0 \leq r \leq m$,*

$$F_p(k) - F_{1-p}(g_p((k - r)_+)) \leq (2r - W)_+. \tag{68}$$

Consequently

$$\lceil 2\Gamma/d \rceil \leq T_m(Q \square P_n) \leq 4 \lceil (M - m)_+/d \rceil + 2. \tag{69}$$

Proof. The potential inequality is trivial if $\Gamma = 0$. Otherwise let $q = (k - r)_+$ and $j = g_p(q)$. If $q \leq Z$, then $k \leq Z + m$ and $2k + p - K \leq 0$, so the current potential is zero. If $q \geq H - Z - 1$, then $j \geq q - 1 \geq H - Z - 2$, and

$$2j + (1 - p) - K \geq 2H - 2Z - 4 - K = \Gamma.$$

The successor potential is therefore Γ . In the remaining region both profiles are on their plateaus, so $j = k - r + \beta + p$. The difference of the unclipped affine expressions is $2r - W$. Common monotone, 1-Lipschitz clipping proves (68).

For two quota shares $r, m - r$, the right sides sum to at most $d = 2m - W$. Both full color classes have potential Γ , giving the lower bound by summing daily decreases. For the upper bound,

start with the minority cohort of size M . Each pair of full-budget days reduces an uncleared cohort by at least d . Thus $2 \lceil (M - m)_+ / d \rceil + 1$ days suffice. Pad early clearing to this fixed odd phase length. The other initial cohort remains its full current color class and, after the odd phase, is also in the minority color with size M . Repeat the phase to obtain the upper bound. $\square$

For a rectangle $Q = P_w$, $w = 2b + 1$, the explicit profiles of Theorem 4.1 permit the sharper corner radius $Z = b^2$ in the preceding argument. At minimum budget, strengthen its potential slightly as follows, for every odd $n \geq w$. Take $K = 2b^2 + 2b + 2$ and $\Gamma = (2H - 4b^2 - 2b - 6)_+$. In the low region the current potential is zero for $r < m$ and at most one for $r = m$; the high-region and plateau arguments are unchanged. This gives the useful uniform estimate

$$\max\{0, 2wn - 2w^2 + 2w - 10\} \leq T_{(w+1)/2}(P_w \square P_n) \leq 2wn - 2w - 2. \quad (70)$$

Thus its leading term is $2wn$ for every fixed odd width, independently of whether a sharper correction has been determined.

Return to the general cylinder, with $Z = W(S + 1)$ and the potential of Lemma 12.4. Assume $H \geq H_*$, so $\Gamma > 0$ and $M > m$. Fix an optimal prefix strategy, using padded full quotas as in Section 11. Let Φ_t be the sum of its two current-color potentials and put

$$\eta_t = d - (\Phi_t - \Phi_{t+1}) \geq 0.$$

These are integers, and the upper bound in (69) gives

$$\sum_t \eta_t = dT_m - 2\Gamma \leq 8Z + 6d + 2 = B. \quad (71)$$

Here $d \lceil (M - m)/d \rceil \leq M - m + d - 1$ accounts for the rounding. Call a day *efficient* if $\eta_t = 0$, and *bad* otherwise. There are at most B bad days.

An efficient day gives all m inspections to one cohort. Indeed, if both positive-part losses in (68) are positive, their sum is $d - W < d$. If only one is positive but neither share is m , it is at most $d - 2$. Thus on an efficient day the active cohort loses exactly d potential and the inactive cohort's potential remains exactly constant.

12.4 From flat potentials to empty or full cohorts

Zero inspections strictly increase every intermediate potential:

$$0 < F_p(k) < \Gamma \implies F_{1-p}(g_p(k)) > F_p(k). \quad (72)$$

For a proper majority prefix, $g_0(k) \geq k$, so its unclipped expression increases by at least one after the color reversal. A full majority prefix has potential Γ and is excluded. For a nonempty minority prefix, $g_1(k) \geq k + 1$, giving the same conclusion. Clipping preserves strict growth while the old value is below Γ .

After an active cohort loses d on an efficient day, its potential is strictly below Γ . If it is still positive, (72) forces it to stay active on the next efficient day. A switch is possible only when its potential reaches zero. It cannot become active again in the same efficient run, since zero cannot lose d and its inactive value must stay constant. Hence each consecutive efficient run has at most two focused blocks, and there are at most $2(B + 1)$ such blocks in total.

Call a day *clean* when all m inspections target one cohort and the other cohort is actually empty or actually its full current color class. Flat potential alone is insufficient for this conclusion. We now bound the transient days needed to reach actual emptiness or fullness.

If a nonempty proper support X lies in one color of a connected bipartite graph without isolated vertices, then $X \subsetneq N^2(X)$. Inclusion follows by backtracking along an edge. Equality would make X closed under two-step paths; connectivity makes the two-step graph connected within each color and would force X to be full. Therefore an uninspected nonempty proper prefix grows by at least one every two days until it is full.

During an efficient focused block the inactive potential is constantly zero or Γ . At zero, its count is at most $Z + m = L - 1$; if nonempty, it cannot remain in that range for more than $2L$ days. At Γ , its count is at least $H - Z - 2$, hence within L of its full class; it becomes full within $2L$ days. Empty and full cohorts remain so under further uninspected moves. Each efficient block therefore contributes at most $2L$ nonclean days. Counting all bad days as well, the entire optimal search has at most

$$\mathcal{E} = B + 4L(B + 1) \quad (73)$$

nonclean days.

12.5 A protected band and its two individual cuts

Every one-day count change has absolute value at most $J = m + 1$: profile surplus is between -1 and $\beta + 1$, and each quota is at most m . Record both source counts from each nonclean day. A count k forbids integer cuts c with $k \in [c - J, c + R + J]$, at most $R + 2J + 1$ cuts. There are at most $2\mathcal{E}$ records. Restrict candidate cuts to

$$A + 2J \leq c \leq H - R - A - 2J - 2. \quad (74)$$

There are $H - R - 2A - 4J - 1$ candidates. By (66), this exceeds $2\mathcal{E}(R + 2J + 1)$. Choose a cut forbidden by no record.

No nonclean transition can touch the protected interval $[c, c + R]$. A cohort in that interval is therefore active on every day, receiving all m inspections; an inactive clean cohort is empty or full, both outside the interval. Its count never increases. Nor can it cross the interval upwards: a nonclean transition cannot touch it, and a clean active transition does not increase. Both cohorts begin above and finish below the interval.

For each *initial* cohort i , choose its first count x_i at most $c + \Delta + J$. The previous count is larger, so the displacement bound gives

$$c + \Delta < x_i \leq c + \Delta + J. \quad (75)$$

The entire protected interval is inside the profile plateaus, including all relevant survivors. A full-budget step there sends k to $k - m + \beta + p$, so every pair of days reduces its count by exactly d and restores its color. The next

$$\tau = 2\Delta/d$$

days therefore take x_i to $x_i - \Delta \in (c, c + J]$. This duration is even, and the trajectory stays in the protected band. Throughout the block its inactive companion remains empty or full. The two crossing blocks cannot overlap.

Define the individual endpoints

$$\ell_i = x_i - \Delta, \quad u_i = x_i.$$

They need not agree for the two cohorts. This avoids an otherwise real residue problem: when $d > 1$, a trajectory may skip a prescribed count, and the two cohorts need not have matching residues. One common protected band supplies two legitimate individual cuts. Every visit to (ℓ_i, u_i) belongs to the chosen crossing block, because active clean trajectories are nonincreasing and nonclean transitions cannot touch the larger band. Boundary stalls at the minimum budget are harmless; the even block duration preserves the needed phase.

12.6 Removing and enlarging the crossing blocks

Proof of Theorem 12.3. First contract the majority size from H to $H - \Delta$ and delete both crossing blocks of length τ . At each remaining state transform initial cohort i by

$$f_i(k) = \begin{cases} k, & k \leq \ell_i, \\ k - \Delta, & k \geq u_i. \end{cases}$$

No retained state lies strictly between these cases. At the ends of a deleted block, u_i and ℓ_i both map to ℓ_i . Its inactive companion is empty or full at both ends; a full class maps to the contracted full class. Since τ is even, the color labels also match. The two state paths therefore glue.

For a low retained source, its survivor is at most ℓ_i . The contracted far-distance satisfies

$$(H - \Delta) - \ell_i = H - u_i \geq H - (c + R) \geq A + 2J + 2,$$

so the low profile identity (63) applies. Its successor cannot enter the removed interval, by the protected-band argument. For a high retained source, the contracted survivor is at least

$$\ell_i - m > c - m > Z.$$

The stable upper profile identity (64) therefore gives

$$g_p^H(q + \Delta) = g_p^{H-\Delta}(q) + \Delta.$$

These identities verify every retained transition. Quotas remain within the same budget, and unused quota may be wasted. The result is a physical prefix search on length $n - 2\Delta/W$. Its majority size is at least $H_* - \Delta \geq 2Z + 6J + 4$, so its length exceeds $4(S + 1)$ and the longitudinal coordinate remains longest. Thus all stable profile identities still apply. The resulting search is completed in $T_m(Q \square P_n) - 4\Delta/d$ days. Hence

$$T_m(Q \square P_{n-2\Delta/W}) \leq T_m(Q \square P_n) - 4\Delta/d. \quad (76)$$

Conversely, start with an optimal search on any $H \geq H_*$ and its protected blocks. Enlarge H to $H + \Delta$. Keep lower counts at or below ℓ_i unchanged and increase upper counts at or above u_i by Δ . Replace each block from u_i to ℓ_i by a clean block from $u_i + \Delta$ to ℓ_i , lasting $4\Delta/d$ days. The affine profiles realize this trajectory explicitly, with the inactive companion empty or the enlarged full class. The extra τ days per block are even. The same low and high identities verify every other transition. Thus

$$T_m(Q \square P_{n+2\Delta/W}) \leq T_m(Q \square P_n) + 4\Delta/d.$$

Apply (76) at $H + \Delta$ for the reverse inequality. Since $\Delta/W = d/g$ and $\Delta/d = W/g$, this proves (67). $\square$

Let n_0 be the least odd length at or beyond the threshold. For each of the d/g odd residue classes modulo $2d/g$, the exact two-count recurrence determines the optimum and an optimal strategy at one of

$$n_0, n_0 + 2, \dots, n_0 + 2d/g - 2.$$

The theorem then determines every later time in that class and constructs its optimal strategy by insertion. In particular,

$$T_m(Q \sqcup P_n) = \frac{2W}{2m - W} n + O_{Q,m}(1).$$

At the eventual minimum budget there is one offset $C(Q)$, with $T_{(W+1)/2}(Q \sqcup P_n) = 2Wn - C(Q)$ for every sufficiently large odd n . The offset can depend on the transverse shape, not only on its number of rooms. For rectangular cross-sections $Q = P_w$, the stronger bounds in (70) give $2w + 2 \leq C(w) \leq 2w^2 - 2w + 10$. Finding smaller corner recurrences or closed expressions for the eventual offsets remains a further problem.

12.7 An exact three-dimensional family

For one cross-section the finite corner calculation gives the optimum from the shortest nontrivial cylinder, without the conservative threshold in Theorem 12.3.

Theorem 12.5. *For every odd $n \geq 3$, five inspections per day are necessary and sufficient on $P_3 \sqcup P_3 \sqcup P_n$, and*

$$T_5(P_3 \sqcup P_3 \sqcup P_n) = 18n - 36.$$

Proof. Put $H = (9n + 1)/2$. For $Q = P_3 \sqcup P_3$, the ranks of bottom-slice vertices in their respective parity orders, and their transverse costs, are

p	ranks	costs
0	(1, 2, 3, 5, 8)	(2, 1, 1, 0, 0)
1	(1, 2, 4, 6)	(2, 1, 1, 0).

These ranks already stabilize at $n = 3$. The last even bottom point $(2, 2, 0)$ is first in weight layer four, and its earlier even layers have longitudinal coordinate at most two. The last odd bottom point $(1, 2, 0)$ is sixth in its parity order: its predecessors are the three unit vectors and $(2, 1, 0)$, $(2, 0, 1)$. Enlarging the longitudinal path therefore introduces no earlier point in either list.

Let $U_p(k)$ be the cumulative listed cost through rank k , and let $B_p(k)$ count the listed ranks at most k . The cost and complement argument of Lemma 12.1 gives, for $k > 0$,

$$g_0(k) = k + U_0(k) - 5 + B_0(H - k), \quad g_1(k) = k + 1 + U_1(k) - 4 + B_1(H - 1 - k),$$

with $g_0(0) = g_1(0) = 0$. Thus the corner radius is eight, and the central profiles are $k + 4$ and $k + 5$. The majority surplus is at most four and equals four at $k = 3$: $U_0(3) = 4$ and $B_0(H - 3) = 5$ for every $H \geq 14$. Corollary 5.6 gives the minimum budget five.

For the exact time use the identical charge sequences

$$c_0 = c_1 = (0, 0, \frac{1}{3}, \frac{2}{3}, 1, 1).$$

They satisfy the daily charge bound (54). The rational shortest-charge potentials described in the preceding section are checked against every transition

$$F_p(k) - F_{1-p}(g_p(\max\{k - r, 0\})) \leq c_p(r), \quad 0 \leq r \leq 5.$$

Together with a minority-first serial strategy, the five finite certificates give

n	3	5	7	9	11
$F_0(H) + F_1(H - 1)$	18	54	90	126	162
strategy length	18	54	90	126	162.

Here the strategy spends all available inspections on the first cohort until it is captured, uses any remaining inspections on its finishing day on the second cohort, and then finishes that cohort. The finite verifier checks the physical room neighborhoods and replays these inspection sets, as well as checking all 1,950 rational inequalities.

At the base $n = 11$, take $H = 50$, $c = 25$, and $D = 6$. The exact arrays satisfy

$$F_p(k) = 2k - 14 + p \quad (13 \leq k \leq 37).$$

This is the full collar $[c - 2D, c + 2D]$, and $c \geq 8 + 2D$, $H - c \geq 8 + 2D + 2$. Both active cohorts visit count 25. At the first visit the other cohort is full, since $25 > 5$ places the visit before the first cohort's finishing day. At the second visit the first cohort is empty.

Consequently Lemmas 11.2 and 11.3, in their compatible-profile formulation, apply with corner radius eight. Increasing H by Δ inserts an affine interval in each potential and raises their initial sum by 4Δ . Inserting 2Δ full-budget days at each serial cut realizes the same increase in time. Each pair of inserted days reduces the active count by one and restores its color, while the other cohort stays full or empty. All remaining transitions follow the stable low and translated high profiles above.

For every odd $n \geq 11$, choose $\Delta = 9(n - 11)/2$. The matching bounds are then $162 + 4\Delta = 18n - 36$. The finite certificates cover the smaller odd lengths, completing the proof. $\square$

The exact verifier and receipt are included in the companion artifact:

```
src/three_by_three_cylinder_time.py
research/three-by-three-cylinder-time.json
```

The receipt stores the corner tables, charges, finite values, and complete base potential arrays. An independent verifier also checked actual room neighborhoods and schedules at seven lengths, including 13 and 31. These checks establish the finite premises of the insertion proof. The variable-length theorem is an ordinary proof with exact rational certificates; it is not claimed here as a complete physical Lean theorem.

13 Higher-dimensional boxes

The preceding compression theorems work in arbitrary dimension, but do not by themselves provide a closed time formula for every box. We give two unbounded applications and a fully certified three-dimensional example.

13.1 A side of length two

Let H be bipartite, with color function χ , and put $G = H \square P_2$. In either color class of G , there is exactly one vertex above each vertex v of H : its second coordinate is $s - \chi(v)$ modulo two. Under this identification,

$$\pi(N_G(R)) = \pi(R) \cup N_H(\pi(R)).$$

The vertical move preserves the projection, and an H -move changes it to an H -neighbor. This proves the equality for arbitrary sets, including boundary and empty sets.

Suppose H has an order with prefixes I_k such that every k -set has at least $c(k)$ vertices in its closed neighborhood and $I_k \cup N_H(I_k) = I_{c(k)}$. Put $A = |H|$. The two initial-color cohorts then have the exact state transition

$$(a, b) \mapsto (c(\max\{a - p, 0\}), c(\max\{b - m + p, 0\})), \quad 0 \leq p \leq m, \quad (77)$$

starting at (A, A) . A state is capturable on the current day exactly when $a + b \leq m$. Tail inspections of the prefixes attain each step. Conversely the cardinalities in every unrestricted search dominate these counters when the same allocations are used. Thus shortest-path search on $(A + 1)^2$ states returns both the exact time and an actual optimal strategy, using $O((m + 1)(A + 1)^2)$ transitions after the profile is known.

Theorem 13.1. *Under the preceding closed-neighborhood nesting hypothesis,*

$$h(H \square P_2) = 1 + \max_{0 \leq k \leq A} (c(k) - k).$$

Both this feasibility formula and recurrence (77) apply to every Cartesian box having a side of length two.

Proof. Write $B = \max_k (c(k) - k)$. If $m \leq B$, select k with $c(k) \geq k + m$. A cohort of size at least $k + m$ retains at least k possible positions after inspection and has at least $k + m$ after movement. Initially $A \geq k + m$. Since $k > 0$, the invariant prevents capture forever. If $m > B$, a canonical cohort of size $b > m$ has next size at most $b - m + B < b$. Clear it, then clear the other cohort in the same way; the untouched projected cohort stays full.

For Cartesian H , omit length-one factors, list the other side lengths in increasing order, and order vertices by increasing coordinate sum, breaking a tie by putting the larger first differing coordinate first. The classical simplicial isoperimetric theorem gives exactly the closed-neighborhood nesting hypothesis. We use the statement and definitions in Otachi–Suda [12, Theorem 2.5], where the result is attributed to Moghadam and Bollobás–Leader. For the one-vertex empty product take $c(0) = 0, c(1) = 1$. $\square$

For $a, b \geq 2$, the vertex boundary width of $P_a \square P_b$ is $\min(a, b)$, so $h(P_a \square P_b \square P_2) = \min(a, b) + 1$. If exactly one of a, b is one, the remaining nontrivial ladder has threshold two; if both are one, it is a single edge with threshold one. The theorem also covers all hypercubes. Its input is a classical *closed*-neighborhood theorem; it does not assert open-neighborhood nesting on arbitrary even rectangles.

13.2 Exact profile transfer for either longitudinal parity

There is a useful further recurrence even when global minimizing prefixes are unavailable. Let a bipartite transverse graph Q have compatible minimizing parity orders with profiles g_0, g_1 . For a one-color support in $Q \square P_n$, let k_j be its size in column $j = 0, \dots, n - 1$, and let p be its global color.

Proposition 13.2. *For each prescribed column-count vector, the exact minimum neighborhood size is*

$$\sum_{j=0}^{n-1} \max\{g_{(p+j) \bmod 2}(k_j), k_{j-1}, k_{j+1}\}, \quad k_{-1} = k_n = 0.$$

This holds for every positive n , including even n .

Proof. In column j , the neighborhood is the union of the transverse neighborhood of that column and the supports in the two adjacent columns. Its size is at least the displayed maximum. Replace every column support by the corresponding transverse prefix. All three sets are now prefixes of the same opposite color order, and their union has size exactly the maximum. These replacements attain all column minima simultaneously. $\square$

Minimize the sum over vectors with $\sum k_j = k$ to obtain the exact global cardinality profile. A transfer state remembers the previous count a , current count b , and cumulative count. Appending c charges $\max\{g(b), a, c\}$ and replaces the pair by (b, c) . Zero counts at the two outside columns give the boundary conditions. Writing $W = |Q|$, this direct recurrence uses $O(n^2 W^4)$ arithmetic operations and $O(nW^3)$ memory once the transverse profiles are known. It applies in particular to every all-odd cross-section.

This computes exact neighborhood minima, not compatible global orders. The minimizing vector can depend on k without being nested. A search using only these cardinality profiles therefore supplies a lower time bound; it is not asserted to attain that bound on arbitrary even cylinders. The accompanying code checks the recurrence against all 524 288 one-color supports of the $3 \times 3 \times 4$ box.

13.3 The complete $3 \times 3 \times 3$ example

Encode a room $(x, y, z) \in \{0, 1, 2\}^3$ by $9x + 3y + z$. Its color is $x + y + z$ modulo two. The two classes have sizes 14, 13. Their minimum open-neighborhood sizes are the following; a dash indicates a source cardinality larger than the class.

Layer $x = 0$	Layer $x = 1$	Layer $x = 2$																											
<table border="1"> <tr><td>0</td><td>1</td><td>2</td></tr> <tr><td>3</td><td>4</td><td>5</td></tr> <tr><td>6</td><td>7</td><td>8</td></tr> </table>	0	1	2	3	4	5	6	7	8	<table border="1"> <tr><td>9</td><td>10</td><td>11</td></tr> <tr><td>12</td><td>13</td><td>14</td></tr> <tr><td>15</td><td>16</td><td>17</td></tr> </table>	9	10	11	12	13	14	15	16	17	<table border="1"> <tr><td>18</td><td>19</td><td>20</td></tr> <tr><td>21</td><td>22</td><td>23</td></tr> <tr><td>24</td><td>25</td><td>26</td></tr> </table>	18	19	20	21	22	23	24	25	26
0	1	2																											
3	4	5																											
6	7	8																											
9	10	11																											
12	13	14																											
15	16	17																											
18	19	20																											
21	22	23																											
24	25	26																											

First day of the optimal 18-day, five-probe schedule.

Figure 5: The three-cube drawn as three layers. Adjacent rooms in a layer are connected; rooms at the same position in consecutive layers are also connected. Shading marks the first five inspections of the schedule.

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$g_E(k)$	0	3	5	7	8	9	10	10	11	12	12	13	13	13	13
$g_O(k)$	0	4	6	7	9	10	11	12	12	13	13	14	14	14	–

Here is a finite certificate procedure for the table. For each color, list its vertices; recursively either include or exclude the next vertex, maintaining the selected count k and the union U of its physical neighbor sets. At every leaf check $|U| \geq g(k)$. This has $2^{14} + 2^{13} = 24576$ leaves. The generic recursion is proved sound in Lean before the concrete finite check is kernel evaluated. Prefixes in increasing coordinate sum with decreasing lexicographic tie breaking attain the values. Neither the lower bound nor the formal classification assumes that an arbitrary search uses these prefixes.

For a fixed budget define the relaxed count successor

$$f_p(a, b) = (g_O(\max\{b - m + p, 0\}), g_E(\max\{a - p, 0\})).$$

All arbitrary physical successors dominate one of these. The finite calculation below independently specifies the time lower bounds. It uses just $15 \cdot 14 = 210$ count pairs; W_t denotes relaxed states that can reach zero in at most t inspection rounds.

```

W = {(0, 0)}
for t = 1, 2, ...:
  Wnext = W union {(a,b): 0<=a<=14, 0<=b<=13,
                      f_p(a,b) belongs to W for some 0<=p<=m}
  if (14,13) belongs to Wnext: return t
  if Wnext == W: return infinity
W = Wnext

```

The required lower endpoint checks are 18, 10, 6, 4, 3 at budgets 5, 6, 8, 11, 12, respectively. In Lean the corresponding distance potentials are stored as finite integers and checked to be nondecreasing in both counts, zero at zero, and to drop by at most one under every relaxed transition. This proves the lower bound without trusting the search routine that found the potentials. For budgets at most four, there is a shorter proof: every same-color triple has at least seven neighbors, so a cohort of at least seven positions can never fall below seven after four inspections and movement.

For clarity, explicit upper schedules are given next. Each row is a sequence of daily inspection sets; a superscript $\times 2$ means repeat the entire listed sequence twice. The physical update $B \leftarrow N(B \setminus S)$, starting from all 27 rooms, verifies every schedule directly.

m	Inspection sets in order
5	$(\{1, 3, 5, 9, 11\}, \{4, 6, 10, 12, 18\}, \{5, 7, 11, 13, 15\},$ $\{8, 10, 12, 14, 16\}, \{9, 11, 13, 15, 17\}, \{10, 12, 14, 16, 18\},$ $\{11, 13, 15, 19, 21\}, \{8, 14, 16, 20, 22\}, \{15, 17, 21, 23, 25\})^{\times 2}$
6	$(\{8, 14, 16, 20, 22, 26\}, \{5, 7, 11, 13, 19, 25\}, \{2, 4, 10, 16, 22, 24\}, \{1, 7, 13, 15, 19, 21\},$ $\{0, 4, 6, 10, 12, 18\})^{\times 2}$
7	$(\{8, 14, 16, 20, 22, 24, 26\}, \{5, 7, 11, 13, 15, 19, 21\}, \{0, 2, 4, 6, 10, 12, 18\})^{\times 2}$
9	$\{2, 4, 8, 14, 16, 20, 22, 24, 26\}, \{1, 3, 7, 9, 11, 13, 15, 19, 21\},$ $\{5, 7, 11, 13, 15, 17, 19, 23, 25\}, \{0, 2, 4, 6, 10, 12, 18, 22, 24\}$
12	$\{2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26\}, \{1, 3, 8, 9, 14, 16, 19, 20, 21, 22, 24, 26\},$ $\{1, 3, 5, 7, 9, 11, 13, 15, 19, 21\}$
13	All odd-numbered rooms, on each of two consecutive days.

Larger budgets inherit these schedules. Fewer than 27 probes cannot capture every possible initial room in one day, and 27 can. We have therefore proved the complete table stated in the introduction.

Its geometry, finite inequalities, physical schedules, and interpretation as capture of every actual target walk are all checked in Lean.

13.4 A uniform height strategy for cylinders

Theorem 13.3. *If H is any finite bipartite graph with $A \geq 1$ vertices and $n \geq 2$, then*

$$h(H \square P_n) \leq \lfloor A/2 \rfloor + 1.$$

For $m > A/2$, at most $2 \lceil A(n+2)/(2m-A) \rceil$ days suffice.

Proof. We first clear one initial-color cohort. Write the path coordinate as $0 \leq z < n$. Bound the possible positions in each fiber by a cutoff a_v , of the current required parity, such that $|a_u - a_v| = 1$ whenever uv is an edge of H . Initially take $a_v = n - 1$ or $n - 2$ according to its parity; this bounds the full cohort. Cutoffs may later be negative or exceed the board.

At a local maximum v , lower a_v by two. Record the room at the old height if it lies on the board. Every adjacent difference stays one. A sequence of m such virtual operations records at most m distinct rooms: repeated operations in a fiber use strictly decreasing heights. Inspect those rooms simultaneously. The survivors lie below the new cutoffs a'_v . After movement their cutoffs are bounded by $a'_v + 1$: a path move increases height by at most one, and an H -move arrives from u with $a'_u \leq a'_v + 1$.

Choose the virtual operations in a fixed cyclic order. List first the initially higher color class of H , then the other class, and repeat this A -letter word. Every operation is at a local maximum; after a whole color class has been lowered, the other is higher. Adding one to all cutoffs after a day changes no comparison. Isolated vertices of H cause no difficulty.

After t days every vertex has been lowered at least $\lfloor mt/A \rfloor$ times, so

$$\max_v a_v(t) \leq n - 1 + t - 2 \lfloor mt/A \rfloor < n + 1 - t(2m - A)/A.$$

The displayed number of days per cohort makes every cutoff negative. Since $H \square P_n$ has no isolated vertices, an empty post-movement belief certifies capture during inspection. Repeat for the other initial cohort, starting with the full bound of its then-current parity. $\square$

Corollary 13.4. *For every $n \geq 3$, $h(P_3 \square P_3 \square P_n) = 5$. The remaining values are two at $n = 1$ and four at $n = 2$.*

Proof. For $n \geq 3$ the board contains the three-cube. An evader may choose to stay in that subgraph, so its hunting number is at least five. Theorem 13.3 gives the upper bound with $A = 9$. At $n = 1$ use the rectangle theorem; at $n = 2$ use Theorem 13.1 with the 3×3 base. $\square$

For a general box this construction can be applied along a longest axis, giving $\lfloor (\prod_{i \neq j} n_i)/2 \rfloor + 1$ as a sufficient budget. It can be loose. In particular the complete time classification of longer $3 \times 3 \times n$ cylinders remains open here.

14 Formal verification, reproducibility, and open problems

The formal development uses Lean 4.33.1 and its standard library. It contains no admitted proof, project-specific mathematical axiom, or trusted external solver result. Finite certificates are reduced

by the Lean kernel. The dependency reports of the final theorems use only the usual logical principles of propositional extensionality, classical choice, and quotient soundness, where applicable.

Three graph classifications are complete in Lean:

- `PathClassification.can_capture_iff` and `exact_minimum` cover every positive path length and budget, including the one-room and one-probe cases.
- `LadderGame` connects the complete two-counter optimum to actual two-row rooms, daily cardinality budgets, and arbitrary avoiding walks, including the single-edge degeneration.
- `ThreeCubeTimeClassification.can_capture_iff` and `exact_minimum` cover every budget on the $3 \times 3 \times 3$ box. The statement quantifies over unrestricted room-valued inspection schedules. It does not assume a preferred order or support shape.

The abstract strategy-comparison theorem is also formalized in `StrategyCompression`. Its concrete odd-path and square operators currently have ordinary proofs. The three-row and four-row classifications, five-row classifications, arbitrary-dimensional odd-box isoperimetry, general side-two application, and unbounded cylinder construction also currently have ordinary proofs, not complete Lean developments. The scalar potentials in the five-row proofs have separate Lean checks, including row-deficit consequences, the four-probe transition and equality lemmas, and the three finite high-budget bounds. The discounted potential for five or more probes and the entire physical-geometric bridge are not yet formalized. Successful checking of the formal files must not be represented as formal verification of those additional theorems.

The generic profile complement identity and the one-unit difference of the maximum neighborhood surpluses for bipartite parts of sizes $M + 1, M$ are fully formalized. The module is `BipartiteProfileDuality`; its inverse arithmetic is in `ProfileInverse`. The profiles are defined as actual attained neighborhood minima over finite subsets; they are not assumed functions. This verifies the structural profile identity used in the odd-box feasibility corollary, while its geometric nesting and the published hunting-number criterion remain ordinary mathematical dependencies. The minimum-budget interval-insertion theorem and its rational certificates also currently have ordinary proofs and independent exact checks, rather than Lean proofs. The uniform minimum-budget scalar inequality is verified in `UniformOddRank`; the eventual-period theorem for odd cylinders and its geometric corner decomposition currently have ordinary proofs.

The possible-position semantics are proved in `BeliefSemantics`: membership is equivalent to the existence of a legal walk avoiding earlier probes. The finite winning recurrence and its greatest losing trap are proved in `CaptureRecurrence`. Formal rounds start at zero, so a last-round index t corresponds to $t + 1$ inspection days. Every public classification has been separately reviewed for this indexing, its physical graph, its global daily budget, and its quantification over all walks.

The companion at <https://angelraychev.com/princess/> provides the presentation, detailed proof text, and formal sources. The research archive includes scripts for independent physical replay, finite lower certificates, exact small-instance searches, and a dated research log. The source-specific verification receipt records the compiler version, source hashes, dependency order, axiom reports, and exit codes. The complete check runs sequentially with one compiler worker. To reproduce it, install the pinned Lean version and run `python3 src/check_lean.py` from the archive root, optionally giving `-lean` followed by the compiler path. No additional package installation is required.

14.1 What remains open in this investigation

The main unresolved target is a comparably concise account of feasibility and minimum time for arbitrary side lengths in arbitrary dimension. The exact algorithms proved here cover substantial unbounded families; they are not a closed classification of every box. Even where an exact recurrence is established, solving it in a useful explicit form remains a distinct problem. Numerical patterns are recorded as conjectures until the required geometry and all parameter ranges are proved.

The arbitrary-dimensional odd-box theorem is established by a uniform exchange argument, extending the two-dimensional base without inferring an unbounded statement from numerical checks. Boxes with even sides greater than two require further ideas: parity-open nesting already fails on some even rectangles. Also, optimal strategies often finish one initial-color cohort before working substantially on the other, with one shared transition day; an unrestricted exchange theorem that would justify this pattern in every case is not presently asserted.

14.2 Research and writing provenance

The path results originated in the author’s October 2019–March 2020 work. The present shorter proofs were reconstructed from the historical manuscripts; the original work’s credit is not re-assigned to the tool used for reconstruction. The 2026 grid and box investigation, new proof organizations, experiments, formal developments, and this draft were developed with substantial assistance from Astra 6 operating through the Codex harness. The research log distinguishes recovered results, known literature, new derivations with unverified priority, failed approaches, and unresolved claims. The human author retains responsibility for the manuscript and any eventual submission.

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